

INFOTACT SOLUTIONS | DATA ANALYTICS INTERNSHIP

CareFlow

Clinical Pathway Process Mining

Project Documentation

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TEAM	Akshar Patel, Prachi Bhilare, Sagar Janathe, Prashanth
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All data in this project is synthetic. No real patient information is used.

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Document Control

Document information

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Status	Draft for mentor review

Purpose of this document

This document describes the CareFlow system as it exists in the four repository branches: what each component does, how the components fit together, what data they consume and produce, how to run them, and how their output was checked. It is written for a reader who needs to understand, operate, verify or extend the project without having taken part in it.

It is documentation, not a narrative report. Each chapter can be read on its own. Design intent, interfaces and procedures are described first; analytical results are included where they are needed to explain what a module produces or to show that it behaves correctly.

Intended audience

Reader	Where to start
Project mentor / reviewer	Chapter 1 (scope), Chapter 12 (validation), Chapter 15 (known issues)
Data analyst joining the project	Chapters 3 to 6 (architecture and data), then Chapters 8 and 9 (modules)
Machine-learning engineer	Chapter 7 (preparation module), Chapter 6 (data dictionary), Chapter 16 (modelling plan)
Operations / hospital manager	Chapter 10 (results), Chapter 11 (dashboards), Chapter 17 (recommendations)
Maintainer	Chapter 13 (operations), Chapter 14 (maintenance), Appendices A and B

Conventions used

- `monospace` text marks file names, column names, commands and code.
- **Module IDs.** M-xx identifies a Python analysis module, N-xx a notebook, R-xx a requirement, I-xx a known issue.
- **Coloured callouts.** NOTE (teal) gives background, DECISION (amber) records a design choice and its reason, WARNING (coral) flags a risk or a discrepancy.
- All durations are in minutes unless stated. "Case" means one patient encounter and "event" one recorded activity within it.

Revision history

Version	Date	Author	Change
1.0	30 Oct 2026	Akshar Patel	First consolidated documentation of the four branches (23 pages).
2.0	30 Oct 2026	Akshar Patel	Expanded edition: requirements, architecture, data dictionary, module specifications, operations guide, issue log, appendices.

Reference documents

#	Document	Used for
1	Project Assignment Letter, ref ITS/DA/1176 (PDF in branch akshar)	Role, duration, objective, scope of work and deliverables
2	README.md in branch main	Consolidated project summary and sprint log
3	README.md in branch akshar	Preprocessing design, benchmarks, validation checks
4	README.md and docs/*.md in branch prachi	Module write-ups for Days 3 to 18
5	README.md, STAKEHOLDER*SUMMARY.md, day18*notes.md, dashboard*p4*notes.md in branch sagar	Notebook inventory, Power BI notes, stakeholder view
6	Source files (Python, notebooks, CSV, JSON, PBIX)	Behaviour verified directly against code and data

1 Introduction

1.1 Project summary

CareFlow is a data-analytics project that reconstructs patient journeys through a hospital Emergency Department (ED) from timestamped event records. It answers practical operational questions: which routes do patients take, how long does each route take, where does time accumulate between steps, how often are steps repeated, and what is known about a patient at arrival that helps predict a long stay.

The data set contains **1,000 cases** and **6,964 events** across **10 activities**, recorded between 1 January and 2 March 2026. It is synthetic, so the project can be shared freely; the methods, however, are the ones that would be applied to real hospital data.

1.2 Problem statement

Hospitals normally report length of stay as a single number per patient. That number hides how a stay is made up. Two patients with the same total time may have waited in very different places: one for a doctor, another for a laboratory report. Without the sequence of steps and the time between them, managers cannot tell which part of the process to change.

Event logs already exist in most hospital information systems because every registration, triage, test and discharge is time-stamped. Process mining turns that log into a map of the actual process. CareFlow applies this approach and adds a data-preparation layer so the same data can later drive predictive models.

1.3 Goals

Goal	Description	Realised by
G1	Obtain a clean, validated event log and case table	Chapters 6, 7; validation scripts
G2	Discover the pathways patients follow and their frequency	Modules M-04, M-05, M-09, M-10, M-16; notebook N-07
G3	Measure journey duration and the time between consecutive steps	Modules M-06, M-11, M-12; notebook N-06
G4	Identify repeated activities and rework loops	Modules M-02, M-05; notebook N-07
G5	Locate bottleneck candidates and recommend what to review	Modules M-11, M-13
G6	Prepare leakage-free features and targets for prediction	Preparation module (Chapter 7)
G7	Present results for managers	Streamlit dashboard, static charts, Power BI (Chapter 11)
G8	Document the work to industry standard	This document

1.4 Scope

In scope

- Data validation, cleaning checks and leakage-free preprocessing of the CareFlow event log and case table.
- Descriptive process mining: activity frequency, transitions, pathway variants, loopbacks, elapsed times.
- Exploratory analysis: demographics, acuity, shift, physician, hour-of-day, outliers, cohorts, feature importance.
- Dashboards: one Streamlit application, three static charts, four Power BI reports.

Out of scope

- Training and deploying a predictive model. The feature store is prepared for it; the models themselves are future work.

- Real-time streaming or integration with a live hospital information system.
- Clinical outcome analysis. The data has no diagnoses, treatments or outcomes beyond timing.
- Conformance checking against a formal clinical guideline model.

1.5 Stakeholders

Stakeholder	Interest	Main outputs used
ED operations manager	Where do patients wait? Which shifts need cover?	Elapsed-time tables, shift and hour analysis, Power BI overview
Clinical lead / charge nurse	Why do patients return to triage?	Loopback analysis, variant comparison
Laboratory and radiology managers	How long do results take?	Transition summaries, percentiles
Data science team	Clean, reproducible features and targets	Model-ready feature store, scaler parameters
Internship mentor	Evidence of method and quality	Validation log, this document

1.6 Assumptions and constraints

ID	Statement
A1	Timestamps are recorded to the minute and refer to the moment each activity is logged. What a timestamp means (start, end, or entry time) is not documented in the data.
A2	The data is synthetic. Patterns found describe this dataset only.
A3	Each case has exactly one department (Emergency Department) and one assigned physician resource.
A4	Activity order within a case is given by `activity_order` and agrees with timestamp order (verified for the sample log).
C1	Project completion window is one month (5 October to 5 November 2026).
C2	Tools are limited to Python, Jupyter, Streamlit and Power BI Desktop.
C3	The raw 1,000-case file is not committed in the `prachi` branch; analysis modules expect it to be copied in.

1.7 Deliverables

Deliverable	Status	Location / note
Source code	Delivered	`akshar/careflow_impute_scale.py`; 18 scripts in `prachi/src`; 14 notebooks in `sagar/notebooks`
Project documentation	Delivered	This document
Project report	Pending	Separate narrative document, not in the reviewed branches
Presentation (PPT)	Pending	Not in the reviewed branches
Model evaluation results	Partly applicable	Imputation benchmark and 16 validation checks. No predictive model is trained yet.
Final demonstration	Delivered	Streamlit app and four Power BI reports

2 Domain Background

This chapter introduces the concepts the rest of the document relies on. A reader familiar with emergency-department operations and process mining can skip to Chapter 3.

2.1 The Emergency Department patient journey

A patient arriving at an ED is registered, assessed by a nurse (triage) who assigns an urgency level, and then seen by a doctor. Depending on the presentation the doctor may order imaging (X-ray), laboratory tests, or neither. Results are reviewed, the patient may be re-assessed, and finally discharged or transferred. The CareFlow log records ten such activities.

Activity	Domain	What it represents in the log
Registration	Administrative	Arrival and demographic capture. Always the first event (`t0`).
Triage	Administrative	Nurse assigns the acuity score. Appears again in rework cases.
Doctor Assessment	Clinical	First physician examination and care plan.
X-Ray	Diagnostic	Imaging examination performed.
Radiology Result	Diagnostic	Radiologist report available.
Documentation Review	Administrative	Chart reconciliation after diagnostics.
Doctor Reassessment	Clinical	Second physician evaluation, only in rework cases.
Lab Collection	Diagnostic	Blood or specimen drawn.
Lab Result	Diagnostic	Laboratory report available.
Discharge	Administrative	Discharge or transfer. Always the last event.

2.2 Triage and acuity (ESI)

Acuity in the data follows the five-level Emergency Severity Index (ESI). Level 1 is the most urgent (immediate life-saving intervention) and level 5 the least. The log stores it as the integer column `acuity`. In this document **critical** means ESI 1 or 2 (170 cases) and **standard** means ESI 3 to 5 (830 cases).

ESI	Category	Typical meaning
1	Resuscitation	Immediate life-saving intervention required
2	Emergent	High-risk presentation; should not wait
3	Urgent	Stable but needs several resources (for example imaging and laboratory)
4	Less urgent	One resource expected
5	Non-urgent	Consultation only, no resources expected

2.3 Process-mining concepts

Concept	Definition used in this project
Event log	A table with at least a case identifier, an activity name and a timestamp for every recorded event.
Case	One patient encounter, identified by `case_id` (for example CF00278).
Trace	The ordered list of activities of one case.
Variant	A distinct trace. CareFlow has six, labelled V1 to V6.
Directly-follows graph (DFG)	A graph whose nodes are activities and whose edges count how often one activity is immediately followed by another.
Transition	An ordered pair of consecutive events within a case. 5,964 transitions of 16 types exist.

Concept	Definition used in this project
Elapsed time	Minutes between the timestamps of the two events of a transition.
Loopback / rework	An activity re-entered after a later stage, here a second Triage, a Documentation Review and a Doctor Reassessment.
Bottleneck candidate	A transition with a comparatively high elapsed time. It is a candidate for investigation, not a proven bottleneck.

2.4 Metric definitions

Metric names are easy to confuse. The table fixes the meaning used throughout this document and states how each is computed.

Metric	Definition	Source column / note
Total duration (LOS)	`case_end` minus `case_start`	`total_duration_min`; mean 107.5, median 105
Registration to triage	Triage time minus registration time	`registration_to_triage_min`; mean 12.6
Door to doctor	Doctor Assessment time minus Registration time	`door_to_doctor_min`; mean 43.4 (equals registration-to-triage plus triage-to-doctor)
Triage to doctor	Doctor Assessment time minus first Triage time	Transition summary; mean 30.7
Assessment to discharge	Discharge time minus Doctor Assessment time	`assessment_to_discharge_min`; mean 64.1
X-ray turnaround	Radiology Result minus X-Ray	`xray_to_result_min`; 0 when no X-ray; mean 22.4 were performed
Lab turnaround	Lab Result minus Lab Collection	`lab_turnaround_min`; 0 when no lab; mean 32.6 were performed
Loopback delay	Duration of the rework segment	`loopback_delay_min`; 0 when no rework; mean 35.1 where it occurs
Loopback rate	Share of cases with `loopback_flag` = 1	23.5% (235 of 1,000)
Ends next day (boarding)	Case ends after midnight of the arrival day	`ends_next_day`; 81 cases

> **NOTE | Elapsed time is not waiting time**
> A transition interval contains both waiting and the activity itself. Because the data does not say whether a timestamp marks the start or the end of an activity, an interval such as Lab Collection to Lab Result can include sample transport, analysis and reporting. Wording in this document therefore uses "elapsed time" and reserves "wait" for cases where the name of the column says so.

2.5 Data leakage in healthcare prediction

A model that predicts length of stay at the moment of arrival may only use information available at that moment. If it were given, say, the number of events in the case, it would be using the future. This is called **data leakage** and it produces impressive but useless accuracy. CareFlow addresses it in two ways: a chronological train/test split (so no model is trained on patients who arrive after those it is tested on) and a model-ready feature store restricted to arrival-time attributes (Chapter 7).

3 Project Organisation

3.1 Team and responsibilities

Member	Branch	Responsibility and contribution
Akshar Patel	`akshar`, `main`	Preprocessing pipeline and ML feature store (`careflow_impute_scale.py`), validation log, consolidated `main` README, project documentation.
Prachi Bhilare	`prachi`	Process-mining analysis modules M-01 to M-18, the static charts, the Streamlit dashboard and the module write-ups under `docs/`.
Sagar Janathe	`sagar`	EDA and audit notebooks N-03 to N-16, four Power BI reports, stakeholder summary and notes.
Prashanth	repository owner (no branch)	Team leader and GitHub collaborator: coordinates the team and manages repository access. He has no analysis or code branch of his own.

3.2 Branching model

Each member works on a personal branch named after them, and `main` holds the consolidated README. The branches were supplied as separate snapshots and do not share an identical folder layout. The table lists what each snapshot contains.

Branch	Size	Content summary
`main`	1 file	README.md with architecture, dataset, pipeline, results, sprint log, recommendations.
`akshar`	11 files	One Python script, eight generated artefacts, README and the assignment letter.
`prachi`	62 files	18 scripts, 17 write-ups and 5 chart/diagram files, 20 processed CSVs, one sample log, README.
`sagar`	42 files	14 notebooks, 16 processed data files, 4 legacy data files, 4 Power BI reports, 4 notes.

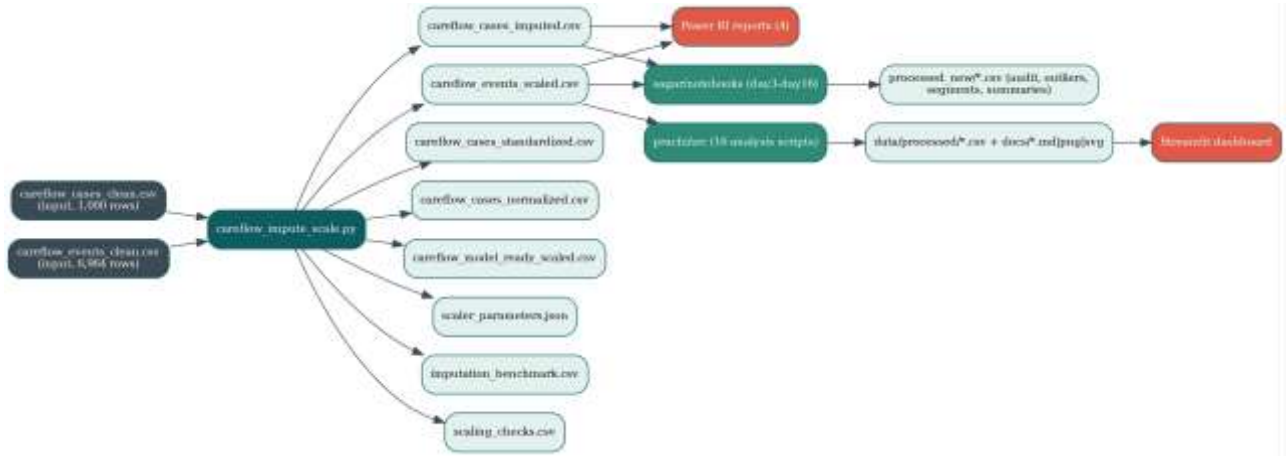


Figure 3.1 File lineage between the preprocessing script, the analysis modules and the dashboards

3.3 Work timeline

Work was organised as a daily sprint. Day numbers differ by branch (see the note below the table), so the table groups activity by phase.

Phase	Akshar	Prachi	Sagar
Setup, validation	Days 1-2: environment, ingestion, timestamp and ID validation	Day 3: validation on 3-case sample	Day 3: schema audit, baseline EDA
Exploration	Days 3-6: EDA, demographics, frequency, duration	Days 4-5: loopbacks, transitions	Days 4-6: demographics, frequency, LOS timeline
Process mining	Days 7-8: trace variants, transition matrix	Days 6-8: workflow discovery, variants, elapsed time	Days 7-8: trace mining, transition heatmap
Analysis, audit	Days 9-10: acuity vs wait, mid-project audit	Days 9-15: frequency, distribution, sequences, bottlenecks, recommendations	Days 9-14: acuity correlation, audit, outliers, cohorts, peak loads, importance
Delivery	Feature store and 16-check log	Days 16-18: charts, Streamlit app, journey comparison	Days 15-18: executive summary, metrics, Power BI, integration notes

NOTE | Day numbering
Prachi and Sagar number days from the start of their own sprint, and Akshar's `main` README lists Days 1 to 10 only. The same "Day 9" is therefore a different task in each branch. Module and notebook IDs in this document (M-xx, N-xx) are used instead of day numbers wherever precision matters.

3.4 Coding and documentation conventions observed

- Python scripts locate the project root with `Path(__file__).resolve().parent.parent` and build input and output paths from it, so they run from any working directory.
- Every analysis script validates required columns, rejects missing values and invalid timestamps, and raises an error rather than continuing silently.
- Each script prints a short summary to the console and writes one CSV into `data/processed/`. A markdown write-up with the same name is placed in `docs/` and contains Objective, Dataset, Methodology, Key Findings, Output Files, Conclusion and Limitation sections.
- Write-ups state limitations explicitly, for example that elapsed time does not equal waiting time.
- Notebooks follow a fixed pattern: imports, load from `data/processed_new/`, analysis cells, a closing markdown cell titled "Key Insights", and an exported summary artefact.

4 Requirements

Requirements were derived from the assignment letter, the objectives in the `main` README and the behaviour of the delivered code. Each requirement carries an identifier so that Chapter 12 can show how it was verified.

4.1 Functional requirements

ID	Requirement	Priority	Implemented in
R-01	The system shall validate that the event log has case, activity and timestamp columns, no missing values, valid timestamps, chronological order and no duplicate rows.	Must	M-01, M-02.M-18 (guard clauses)
R-02	The system shall list every activity with its event count, the number of cases it appears in and the average events per case.	Must	M-07, M-08
R-03	The system shall discover the directly-follows graph of the log and export it as an image.	Must	M-04
R-04	The system shall identify all distinct pathways, their case counts and their average duration.	Must	M-05, M-09, M-16
R-05	The system shall compute elapsed time (mean, median, maximum) for every transition type.	Must	M-06, M-11, M-12
R-06	The system shall identify repeated activities and loopback events and quantify their effect on duration.	Must	M-02, M-16, N-07
R-07	The system shall rank transitions as bottleneck candidates without asserting causes.	Should	M-11, M-13
R-08	The system shall produce a chronological train/test split and fit all statistics on training rows only.	Must	Preparation module
R-09	The system shall distinguish structural from genuine missingness and handle each appropriately.	Must	Preparation module
R-10	The system shall produce standardised and normalised datasets and persist the scaling parameters.	Must	Preparation module
R-11	The system shall produce a feature store containing arrival-time information only, with targets in separate columns.	Must	Preparation module
R-12	The system shall segment performance by acuity, shift, hour, physician and age group.	Should	N-04, N-12, N-13; Section 10
R-13	The system shall detect outliers in duration and flag them.	Should	N-11
R-14	The system shall present key results in an interactive dashboard and static charts.	Must	M-14, M-15, Power BI
R-15	The system shall write its own validation log recording pass or fail for each preprocessing check.	Should	Preparation module

4.2 Non-functional requirements

ID	Attribute	Requirement
NFR-1	Reproducibility	Random operations use fixed seeds (`random_state=0`, `default_rng(42)`); the same input produces the same output.
NFR-2	Leakage safety	No statistic used for imputation or scaling may be computed from test-period rows.
NFR-3	Fail-fast	Scripts raise an explicit error on missing columns, nulls or invalid timestamps instead of producing partial output.
NFR-4	Portability	Paths are resolved relative to the script; no absolute paths are required.
NFR-5	Traceability	Every output CSV can be traced to one script and one input file (Figure 3.1).
NFR-6	Honest interpretation	Documents state limitations: elapsed time is not waiting time; recommendations are for investigation.
NFR-7	Privacy	The dataset is synthetic and contains no direct patient identifiers.
NFR-8	Performance	The full analysis of 6,964 events completes in seconds on a laptop; no distributed tooling is needed.

4.3 Data requirements

Requirement	Detail
Minimum event-log schema	`case_id`, `activity_name` (or `activity`), `timestamp`. Most modules also require `activity_order`.
Preferred case attributes	`age`, `acuity`, `resource`, `department`, `case_start`, `case_end`, and the interval columns listed in Chapter 6.
Timestamp format	Parseable by `pandas.to_datetime`; minute resolution is sufficient.
Ordering	Events within a case must be orderable by `activity_order` or timestamp.
Volume	Tested at 1,000 cases and 6,964 events; larger logs should work but have not been tried.

5 System Architecture

5.1 Architectural overview

CareFlow is a file-based analytics system. There is no database or server: each stage reads CSV or JSON files and writes new ones, so every intermediate result can be opened and inspected. This keeps the project transparent and easy to reproduce, at the price of manual coordination between stages (Section 13.2).

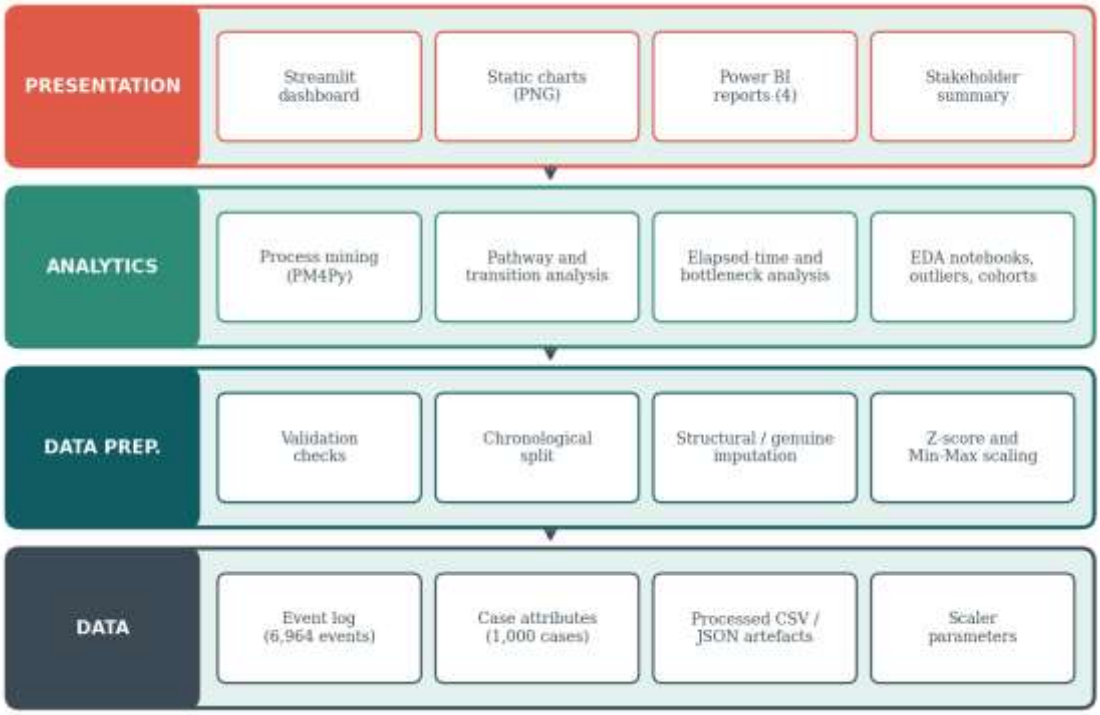


Figure 5.1 Layered architecture of CareFlow



Figure 5.2 End-to-end processing pipeline

5.2 Component responsibilities

Component	Owner / location	Responsibility
Data validation	`prachi/src/validate_event_log.py` and guard clauses in every script	Reject malformed logs before analysis
Preparation module	`akshar/careflow_impute_scale.py`	Split, impute, scale, engineer features, write artefacts and validation log
Process-mining modules	`prachi/src/*.py`	Frequency, transitions, variants, loopbacks, elapsed times, recommendations
Analysis notebooks	`sagar/notebooks/*.ipynb`	Exploratory analysis, audit, outliers, cohorts, importance
Presentation layer	Streamlit app, PNG charts, Power BI	Present results to managers
Artefact store	`data/processed*/` folders	Persist all intermediate and final tables

5.3 Data flow

1. The cleaned case table and event table enter the preparation module together with the two directory arguments.
2. The module writes eight artefacts. `careflow_events_scaled.csv` is the event log used by every process-mining script; `careflow_cases_imputed.csv` is used by the notebooks and Power BI.
3. Each process-mining script reads the event log, derives a summary and writes one CSV to `data/processed/` plus a markdown write-up in `docs/`.
4. The Streamlit dashboard and the chart script read only the summary CSVs, never the raw log.
5. Notebooks read from `data/processed_new/` and save their own summary artefacts there.

5.4 Technology stack

Layer	Technology	Role
Language	Python 3.10 or later	All scripts and notebooks (3.11 used)
Data handling	pandas, NumPy	Loading, grouping, shifting timestamps, aggregation
Process mining	PM4Py	Event-log conversion and DFG discovery (`patient_workflow_discovery.py`)
Machine learning	scikit-learn	IterativeImputer, KNNImputer, SimpleImputer, StandardScaler, MinMaxScaler, RandomForestRegressor
Static visualisation	Matplotlib, Seaborn	Charts and heatmaps in scripts and notebooks
Interactive visualisation	Streamlit, Plotly	Dashboard application and interactive charts
Business intelligence	Power BI Desktop	Four `.pbix` reports
Notebook environment	Jupyter	Exploratory analysis
Version control	Git, GitHub	Branch per member

5.5 Repository layout

The three working branches use different layouts. The trees below show the structure each one has.

Branch akshar

```
akshar/
  careflow_impute_scale.py           preprocessing + validation (180 lines)
  careflow_cases_imputed.csv         1000 x 32
  careflow_cases_standardized.csv    1000 x 33
  careflow_cases_normalized.csv      1000 x 33
  careflow_events_scaled.csv         6964 x 27
  careflow_model_ready_scaled.csv    1000 x 41
  imputation_benchmark.csv  scaling_checks.csv  scaler_parameters.json
  project assignment letter.pdf      README.md
```

Branch prachi

```
prachi/
  src/          18 Python scripts (M-01 .. M-18)
  docs/         17 markdown write-ups, 3 PNG charts, 2 SVG process maps
  data/raw/     sample_event_log.csv (3-case prototype)
  data/processed/   20 summary CSVs
  README.md
```

Branch sagar

```
sagar/
  notebooks/        14 notebooks (day3 .. day16)
  data/processed_new/   16 files (cases, events, outliers, segments, summaries)
  data/processed_old/   4 legacy files (cleaned_careflow.csv, ...)
  dashboard/        4 Power BI reports (.pbix) + dashboard_p4_notes.md
  STAKEHOLDER_SUMMARY.md   day18_notes.md   README.md
```

5.6 Design principles

- **One script, one question.** Each process-mining module answers a single question and writes a single table. This makes outputs easy to cite and to test.
- **Summaries, not raw data, feed dashboards.** The Streamlit app loads small CSVs, so it starts instantly and cannot expose row-level data.
- **Explicit missingness.** A step that did not occur is stored as 0 together with an indicator flag, never as an invented value.
- **Time-respecting statistics.** Anything fitted on data is fitted on the earlier 80% only.
- **Claims limited to evidence.** Recommendations are phrased as items to investigate.

6 Data Design

6.1 Data source and profile

Property	Value
Source	Synthetic ED event log generated for the Infotact internship
Period (case starts)	2026-01-01 09:13 to 2026-03-02 06:58 (61 calendar days)
Cases / events / activities	1,000 / 6,964 / 10
Events per case	4 to 11, mean 6.96 (see distribution below)
Cases per day	mean 16.39, minimum 5, maximum 25
Physicians	12 (DR01 to DR12), 64 to 106 cases each
Patient age	18 to 90, mean 55.0, standard deviation 21.7
Acuity mix	ESI 1: 26, ESI 2: 144, ESI 3: 429, ESI 4: 309, ESI 5: 92
Weekend share	29.7% of cases
Duration (min)	P5 37, median 105, P90 165, P99 208; mean 107.5; skewness 0.189

Events per case

Events	4	6	7	9	11
Cases	267	149	216	279	89

The remaining counts are. Six events is the most common length, produced by the radiology-only and laboratory-only pathways.

6.2 Logical data model

Two tables are linked by `case_id`. The case table has one row per patient and carries attributes and interval summaries. The event table has one row per recorded activity and repeats selected case attributes so that event-level analysis does not need a join.

Relationship	Cardinality	Notes
Case to Event	one to many (1 : 4..11)	Joined on `case_id`; `activity_order` gives sequence
Case to Physician	many to one	Stored as `resource`; 12 distinct values
Case to Department	many to one	Only one value exists
Case to Variant	many to one	`variant_id` V1 to V6
Event to Activity	many to one	10 distinct `activity_name`, grouped by `activity_category`

6.3 Data dictionary: case table

Source: `careflow_cases_imputed.csv`, 1,000 rows, 32 columns. "Observed" shows the range for numbers or the most frequent values for text.

Column	Type	Non-null	Unique	Observed values / range	Description
`case_id`	text	1,000	1,000	CF00278	Unique case key (CF + 5 digits)
`case_start`	datetime	1,000	997	2026-01-15 10:03:00	Timestamp of Registration
`case_end`	datetime	1,000	994	2026-01-24 06:31:00	Timestamp of Discharge
`total_duration_min`	float	1,000	192	12 to 234	case*end minus case*start (length of stay); primary prediction target

Column	Type	Non-null	Unique	Observed values / range	Description
n_events	int	1,000	5	4 to 11	Number of events in the case
n_unique_activities	int	1,000	6	4 to 10	Number of distinct activities in the case
variant_id	text	1,000	6	V1, V2, V3, V4, V5, V6	Pathway label V1 to V6
variant	text	1,000	6	Registration > Triage > Doctor Assessment > Discharge...	Full activity sequence joined with " > "
age	int	1,000	73	18 to 90	Patient age in years
age_group	text	1,000	3	60+, 18-39, 40-59	Band: 18-39, 40-59, 60+
acuity	int	1,000	5	1 to 5	ESI triage level, 1 (urgent) to 5
resource	text	1,000	12	DR07	Assigned physician DR01 to DR12
department	text	1,000	1	Emergency Department	Always Emergency Department
has_xray	int	1,000	2	0 to 1	1 if an X-ray was performed
has_lab	int	1,000	2	0 to 1	1 if laboratory work was performed
has_reassessment	int	1,000	2	0 to 1	1 if Doctor Reassessment occurred
loopback_flag	int	1,000	2	0 to 1	1 if the case contains a rework loop
loopback_reason	text	1,000	2	No loopback, Missing intake paperwork	Recorded reason; "Missing intake paperwork" for all 235 rework cases
registration_to_triage_min	float	1,000	25	1 to 25	Minutes from Registration to first Triage
door_to_doctor_min	float	1,000	81	4 to 84	Minutes from Registration to Doctor Assessment
xray_to_result_min	float	1,000	27	0 to 35	X-Ray to Radiology Result; 0 when no X-ray
lab_turnaround_min	float	1,000	37	0 to 50	Lab Collection to Lab Result; 0 when no lab
assessment_to_discharge_min	float	1,000	146	5 to 176	Doctor Assessment to Discharge
loopback_delay_min	float	1,000	42	0 to 57	Length of rework segment; 0 when none
start_hour	int	1,000	24	0 to 23	Hour of arrival, 0-23
start_weekday	text	1,000	7	Saturday	Weekday name of arrival
start_month	int	1,000	3	1 to 3	Month of arrival (1-3)
is_weekend	bool	1,000	2	True 297 / False 703	True for Saturday or Sunday
shift	text	1,000	3	Evening, Night, Day	Day 07-15, Evening 15-23, Night 23-07
ends_next_day	bool	1,000	2	True 81 / False 919	True if case ends after midnight of arrival day
duration_outlier_flag	bool	1,000	2	True 1 / False 999	True if flagged as a duration outlier (1 case)
split	text	1,000	2	train, test	train (first 80%) or test (last 20%)

6.4 Data dictionary: event table

Source: careflow_events_scaled.csv, 6,964 rows, 27 columns.

Column	Type	Non-null	Unique	Observed values / range	Description
case_id	text	6,964	1,000	CF00005	Case key (foreign key to case table)
activity_name	text	6,964	10	Triage	One of the 10 activities
activity_category	text	6,964	3	Administrative, Diagnostic, Clinical	Administrative, Clinical or Diagnostic
timestamp	datetime	6,964	6,721	2026-02-16 18:14:00	Time the event was logged (minute resolution)
activity_order	int	6,964	11	1 to 11	Position of the event within its case (1 = first)
activity_occurrence	int	6,964	2	1 to 2	How many times this activity has occurred so far in the case
is_repeat_activity	bool	6,964	2	True 235 / False 6729	True when occurrence is 2 or more
minutes_since_prev	float	6,964	61	0 to 60	Minutes since the previous event (0 for the first)
minutes_since_case_start	float	6,964	218	0 to 234	Minutes since Registration

Column	Type	Non-null	Unique	Observed values / range	Description
resource	text	6,964	12	DR07	Physician DR01 to DR12
age	int	6,964	73	18 to 90	Patient age
age_group	text	6,964	3	60+, 18-39, 40-59	Age band
acuity	int	6,964	5	1 to 5	ESI level 1-5
department	text	6,964	1	Emergency Department	Emergency Department
is_loopback_event	bool	6,964	2	True 705 / False 6259	True for events that belong to a rework loop
loopback_reason	text	6,964	2	No loopback, Missing intake paperwork	Reason on rework events, otherwise empty
hour	int	6,964	24	0 to 23	Hour of the event
day_of_week	text	6,964	7	Saturday	Weekday of the event
split	text	6,964	2	train, test	train or test label inherited from the case
minutes_since_prev_z	float	6,964	61	-1.2093 to 3.5138	Z-score of minutes*since*prev
minutes_since_case_start_z	float	6,964	218	-1.25739 to 3.55799	Z-score of minutes*since*case_start
age_z	float	6,964	73	-1.69306 to 1.65809	Z-score of age
acuity_z	float	6,964	5	-2.49903 to 1.83885	Z-score of acuity
minutes_since_prev_mm	float	6,964	61	0 to 1	Min-Max of minutes*since*prev
minutes_since_case_start_mm	float	6,964	218	0 to 1	Min-Max of minutes*since*case_start
age_mm	float	6,964	73	0 to 1	Min-Max of age
acuity_mm	float	6,964	5	0 to 1	Min-Max of acuity

6.5 Data dictionary: model-ready feature store

Source: careflow_model_ready_scaled.csv, 1,000 rows, 41 columns. Only information available at arrival is present among the feature columns. The three target_* columns must never be used as inputs.

Column	Type	Non-null	Unique	Observed values / range	Description
case_id	text	1,000	1,000	CF00278	Case key
case_start	datetime	1,000	997	2026-01-15 10:03:00	Arrival timestamp (not a feature; used for ordering)
age	int	1,000	73	18 to 90	Raw value known at arrival
acuity	int	1,000	5	1 to 5	Raw value known at arrival
start_hour_sin	float	1,000	21	-1 to 1	sin(2*pi*hour/24)
start_hour_cos	float	1,000	22	-1 to 1	cos(2*pi*hour/24)
is_weekend	int	1,000	2	0 to 1	Weekend indicator
age_group_18-39	int	1,000	2	0 to 1	One-hot age band
age_group_40-59	int	1,000	2	0 to 1	One-hot age band
age_group_60+	int	1,000	2	0 to 1	One-hot age band
resource_DR01	int	1,000	2	0 to 1	One-hot physician
resource_DR02	int	1,000	2	0 to 1	One-hot physician
resource_DR03	int	1,000	2	0 to 1	One-hot physician
resource_DR04	int	1,000	2	0 to 1	One-hot physician
resource_DR05	int	1,000	2	0 to 1	One-hot physician
resource_DR06	int	1,000	2	0 to 1	One-hot physician
resource_DR07	int	1,000	2	0 to 1	One-hot physician
resource_DR08	int	1,000	2	0 to 1	One-hot physician
resource_DR09	int	1,000	2	0 to 1	One-hot physician
resource_DR10	int	1,000	2	0 to 1	One-hot physician
resource_DR11	int	1,000	2	0 to 1	One-hot physician

Column	Type	Non-null	Unique	Observed values / range	Description
resource_DR12	int	1,000	2	0 to 1	One-hot physician
shift_Day	int	1,000	2	0 to 1	One-hot shift
shift_Evening	int	1,000	2	0 to 1	One-hot shift
shift_Night	int	1,000	2	0 to 1	One-hot shift
start_weekday_Friday	int	1,000	2	0 to 1	One-hot weekday
start_weekday_Monday	int	1,000	2	0 to 1	One-hot weekday
start_weekday_Saturday	int	1,000	2	0 to 1	One-hot weekday
start_weekday_Sunday	int	1,000	2	0 to 1	One-hot weekday
start_weekday_Thursday	int	1,000	2	0 to 1	One-hot weekday
start_weekday_Tuesday	int	1,000	2	0 to 1	One-hot weekday
start_weekday_Wednesday	int	1,000	2	0 to 1	One-hot weekday
target_total_duration_min	float	1,000	192	12 to 234	TARGET: length of stay (min)
target_loopback	int	1,000	2	0 to 1	TARGET: rework occurred
target_ends_next_day	int	1,000	2	0 to 1	TARGET: case crosses midnight
split	text	1,000	2	train, test	train / test
age_z	float	1,000	73	-1.71401 to 1.646	Z-score (training statistics)
acuity_z	float	1,000	5	-2.51203 to 1.86147	Z-score (training statistics)
age_mm	float	1,000	73	0 to 1	Min-Max (training range)
acuity_mm	float	1,000	5	0 to 1	Min-Max (training range)
target_total_duration_min_z	float	1,000	192	-2.15705 to 2.8837	Z-score (training statistics)

NOTE | One-hot columns are fully expanded
All seven weekdays, all three shifts and all three age groups are present, so each group sums to 1 and is perfectly collinear. Tree-based models are unaffected. Linear or logistic models should drop one column per group (or use a regularised fit).

6.6 Completeness and validity

Check	Result	Evidence
Missing values in case table (after imputation)	0	scaling_checks.csv row 5 and 6
Genuinely missing observed values	0	Pipeline detected found=0 before imputation
Structural gaps	3 columns	xray, lab and loopback-delay intervals are empty by design when the step did not occur
Duplicate event rows	0	Checked by M-01 on the 3-case sample log only; the reviewed code has no dedicated duplicate check for the full log
Invalid timestamps	0	All modules abort if any timestamp fails to parse
Negative intervals	0	Modules drop transitions with negative elapsed time; none were dropped (5,964 of 5,964 kept)
Chronological order within case	Pass	Sorted by activity_order; elapsed times are all non-negative

6.7 Data lineage

Artefact	Produced by	Consumed by
`careflow_cases_imputed.csv`	Preparation module	Notebooks, Power BI, `main` README figures
`careflow_cases_standardized.csv`, `_normalized.csv`	Preparation module	Model training (future)
`careflow_events_scaled.csv`	Preparation module	All M-modules, notebooks, Power BI
`careflow_model_ready_scaled.csv`	Preparation module	Model training (future)
`scaler_parameters.json`	Preparation module	Inference-time transformation
`data/processed/*.csv` (20 files)	Modules M-02 to M-18	Dashboard, chart script, write-ups
`data/processed_new/*` (16 files)	Notebooks N-03 to N-16	Notebooks, stakeholder summary

7 Data Preparation Module

This chapter specifies `careflow_impute_scale.py` (branch `akshar`). It is the only component that writes the shared datasets, so every other module depends on its output.

7.1 Module summary

PREP-01	careflow_impute_scale.py
Purpose	Turn the cleaned case and event tables into leakage-free, scaled, model-ready datasets and log 16 automated checks on its own results.
Inputs	`careflow_cases_clean.csv` (1,000 rows) and `careflow_events_clean.csv` (6,964 rows) in the source directory. Produced by an upstream cleaning step (`careflow_clean.py`) that is not part of the reviewed branches.
Arguments	`argv[1]` source directory (default: current directory), `argv[2]` output directory (default: current directory).
Outputs	Eight files: cases*imputed, cases*standardized, cases*normalized, events*scaled, model*ready*scaled, imputation*benchmark, scaling*checks, scaler_parameters.json.
Dependencies	pandas, NumPy, scikit-learn (IterativeImputer, KNNImputer, SimpleImputer, StandardScaler, MinMaxScaler).
Determinism	`random_state=0` for the imputer, `default_rng(42)` for the benchmark mask. Results are repeatable.
Run time	A few seconds.
Exit behaviour	Prints the 16-check log and a PASS/FAIL count. It does not stop on a failed check; the log must be read.

7.2 Processing steps

The script is organised in five numbered sections that execute in order.

Step	Section	What happens
1	Chronological split	Cases are sorted by `case_start` (stable sort). The 80th percentile of start time becomes the cutoff; cases starting at or before it are `train`, the rest `test`.
2a	Structural missingness	For each of `xray_to_result_min`, `lab_turnaround_min`, `loopback_delay_min`, rows whose indicator flag is 0 are set to NaN to make the gap explicit, checked, then filled with 0.0.
2b	Genuine missingness	If any of eight observed numeric columns has missing values, an IterativeImputer (max_iter=20) is fitted on training rows and applied to all rows. None were found, so this path is dormant.
2c	Benchmark	10% of known values are masked at random, imputed by four methods fitted on training rows, and scored by normalised RMSE against the true values.
3	Case-level scaling	StandardScaler and MinMaxScaler are fitted on training rows for 11 features plus the target, then applied to all 1,000 rows. Parameters are saved.
4	Event-level scaling	Four event features are scaled with scalers fitted on training events (events whose case is in the training set).
5	Feature store	Arrival-time features are assembled, categorical fields one-hot encoded, hour encoded as sine/cosine, targets appended, age and acuity scaled, target scaled separately.
6	Write	Datetimes are formatted as `%Y-%m-%d %H:%M:%S`; CSVs, the JSON file and the check log are written.

7.3 Step 1: chronological split

A random split would place patients from February in the training set and patients from January in the test set. A model could then use information about the future of the process to predict its past. The module avoids this by cutting along time.

Split logic (excerpt)

```
cases = cases.sort_values("case_start", kind="stable").reset_index(drop=True)
cutoff = cases["case_start"].quantile(0.8)
cases["split"] = np.where(cases["case_start"] <= cutoff, "train", "test")
```

Cohort	Cases	Mean stay	Loopback	Mean acuity	Mean age
train (2026-01-01 to 2026-02-18)	800	107.0	23.6%	3.3	54.7
test (2026-02-18 to 2026-03-02)	200	109.5	23%	3.29	56.1

Table 7.1 Train and test cohorts. Cutoff timestamp 2026-02-18 11:00:00

The two cohorts are similar in mean stay (107.0 and 109.5 minutes), rework rate (23.6% and 23.0%) and case mix, so a model evaluated on the later period is not being tested on a markedly different population. All six pathway variants appear in both cohorts (for example V6: 72 training, 17 test).

7.4 Step 2: missingness

Structural missingness

Some interval fields are empty simply because a step did not take place. A patient with no X-ray has no X-ray turnaround. Mean or median imputation would invent a procedure that never happened and would contaminate the feature with a false average. The module therefore stores 0.0 and relies on the paired indicator column so a model can separate "no X-ray" from "very fast X-ray".

Interval field	Indicator flag	Cases with value > 0 / mean when performed
`xray_to_result_min`	`has_xray`	584 cases; mean 22.4 min, median 22, P90 33
`lab_turnaround_min`	`has_lab`	371 cases; mean 32.6 min, median 32, P90 48
`loopback_delay_min`	`has_reassessment`	235 cases; mean 35.1 min, median 35, P90 47

DECISION | Zero plus flag rather than imputation

A zero in an interval column means "step absent" only because the flag is 0. Reports computed over the whole column therefore understate the typical duration. For example the column mean of `xray_to_result_min` is 13.1 minutes, but the mean among patients who actually had an X-ray is 22.4 minutes. Always filter on the flag before summarising these three columns.

Genuine missingness

Eight observed columns were examined for random gaps: `age`, `acuity`, `n_events`, `total_duration_min`, `registration_to_triage_min`, `door_to_doctor_min`, `assessment_to_discharge_min` and `start_hour`. None had any (`found=0`), so no observed value was altered. The code path is retained so that future data with gaps is handled correctly without a rewrite.

7.5 Imputation benchmark

Because the method choice should rest on evidence, the module measures it. For each of the eight columns, 10% of known values are hidden, filled by each method, and compared with the hidden truth. Statistics are fitted on training rows that were not masked. The error is the RMSE divided by the column's standard deviation, so that columns on different scales can be averaged.

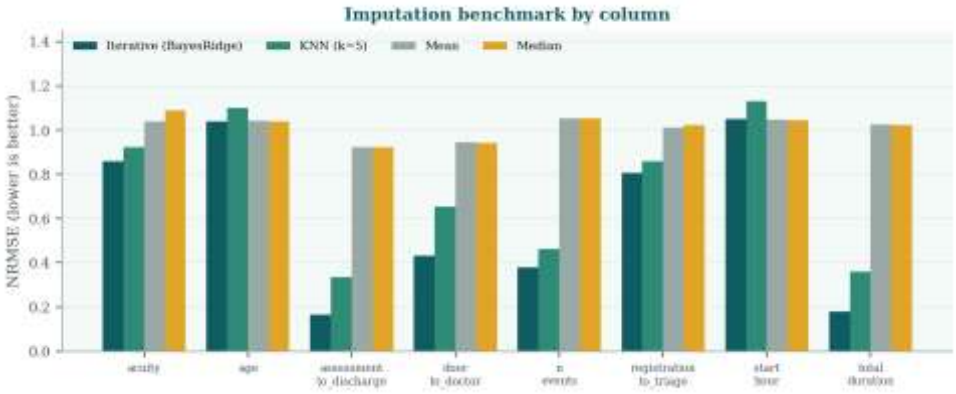


Figure 7.1 Normalised RMSE by column and imputation method (lower is better)

Column	Iterative (BayesRidge)	KNN (k=5)	Mean	Median
acuity	0.86	0.92	1.04	1.09
age	1.04	1.10	1.04	1.04
assessment*to*discharge_min	0.17	0.33	0.92	0.92
door*to*doctor_min	0.43	0.65	0.95	0.94
n_events	0.38	0.46	1.05	1.05
registration*to*triage_min	0.81	0.86	1.01	1.02
start_hour	1.05	1.13	1.05	1.04
total*duration*min	0.18	0.36	1.02	1.02
Mean	0.61	0.73	1.01	1.02

Table 7.2 Imputation benchmark results

Iterative imputation with Bayesian Ridge regression has the lowest mean error (0.61) because it exploits relationships between interval columns: duration columns are strongly linked, so a missing `total_duration_min` can be reconstructed from the others (NRMSE 0.18 against 1.02 for the median). For `age` and `start_hour`, which are unrelated to other columns in this synthetic data, no method beats the baseline and the median is marginally best. This is the expected behaviour for independent variables and not a defect.

7.6 Step 3 and 4: scaling

Scaling puts features with different units on a comparable footing, which matters for distance-based and gradient-based learners. Two variants are produced because models differ in preference.

Variant	Formula	Typical use
Z-score (`_z`)	z = (x - mean*train) / std*train	Linear and logistic regression, SVM, neural networks
Min-Max (`_mm`)	m = (x - min*train) / (max*train - min_train)	KNN, neural networks with bounded inputs, visualisation



Figure 7.2 Effect of scaling on door_to_doctor_min. The shape is unchanged; only the axis differs

Case level. Eleven features plus the target are scaled: `age`, `acuity`, `n_events`, `n_unique_activities`, `registration_to_triage_min`, `door_to_doctor_min`, `xray_to_result_min`, `lab_turnaround_min`, `assessment_to_discharge_min`, `loopback_delay_min`, `start_hour`, and `total_duration_min`. In the standardised and normalised outputs the raw feature columns are replaced by their scaled versions, and the raw target is kept.

Event level. `minutes_since_prev`, `minutes_since_case_start`, `age` and `acuity` are scaled over the 6,964 events, with scalers fitted on events belonging to training cases. Both the raw and the scaled columns are kept in `careflow_events_scaled.csv`.

Out-of-range test values. Min-Max scaling is not clipped. Two test-period values fall outside [0, 1] (for instance a test door-to-doctor value larger than any training value). Clipping would hide the fact that the model is extrapolating, so it is left to the consumer to decide.

Scaler parameter file

`scaler_parameters.json` stores everything needed to transform a new patient exactly as the training data was transformed. The top-level keys are:

Key	Content
`split_cutoff`	Cutoff timestamp string
`fit_on`	The text "train rows only"
`case_level`	For each of 12 columns: mean, std, min, max
`event_level`	Same four statistics for the four event features
`model_ready`	Statistics for `age`, `acuity` and the duration target (mean and std)

Case-level feature	Mean	Std	Min	Max
`age`	54.73	21.43	18.00	90.00
`acuity`	3.30	0.92	1.00	5.00
`n_events`	6.96	2.28	4.00	11.00
`n_unique_activities`	6.73	2.01	4.00	10.00
`registration_to_triage_min`	12.60	6.76	1.00	25.00
`door_to_doctor_min`	43.01	18.94	4.00	84.00
`xray_to_result_min`	12.96	12.39	0.00	35.00
`lab_turnaround_min`	12.05	16.97	0.00	50.00
`assessment_to_discharge_min`	63.99	39.41	5.00	176.00
`loopback_delay_min`	8.35	15.63	0.00	57.00
`start_hour`	11.55	6.92	0.00	23.00
`total_duration_min`	107.00	44.04	15.00	234.00

Table 7.3 Training-set parameters stored in scaler_parameters.json

7.7 Step 5: model-ready feature store

The feature store contains one row per case and is designed for prediction at arrival. It deliberately excludes every column that is only known after the patient has been seen.

Group	Columns	Transformation
Demographics	`age`, `age_z`, `age_mm`	Raw, Z, Min-Max
Age band	`age_group_18-39`, `_40-59`, `_60+`	One-hot
Severity	`acuity`, `acuity_z`, `acuity_mm`	ESI 1-5 in three scales
Arrival hour	`start_hour_sin`, `start_hour_cos`	sin and cos of 2*pi*hour/24
Calendar	`is_weekend`, 7 `start_weekday_*`	Integer flag, one-hot
Shift	`shift_Day`, `shift_Evening`, `shift_Night`	One-hot
Physician	`resource_DR01` to `resource_DR12`	One-hot
Targets	`target_total_duration_min`, `target_total_duration_min_z`, `target_loopback`, `target_ends_next_day`	Regression target (raw and standardised), two binary targets
Housekeeping	`case_id`, `case_start`, `split`	Key, ordering, train/test label

> **NOTE | Why sine and cosine for the hour**
> Hour 23 and hour 0 are one hour apart, but as plain numbers they are 23 apart. Encoding the hour on a circle with sine and cosine gives the model a correct sense of distance across midnight, which is exactly the boundary that matters for the boarding target.

Target balance. Loopback is positive in 235 of 1,000 cases (23.5%) and ends-next-day in 81 (8.1%). Accuracy alone is misleading for such classes; use precision-recall AUC, balanced accuracy, recall at fixed precision and the Brier score. The duration target is right-skewed (skewness 0.189), so Huber loss or median-based metrics (MAE) are more robust than squared error.

7.8 Validation checks

The module logs its own checks, each recorded as PASS or FAIL with a detail string. They fall into four groups: split integrity, missingness handling, scaling correctness and structural consistency. All 16 pass in the supplied log.

#	Check	Result	Purpose / detail
1	split is chronological (max train start <= min test start)	PASS	Guards against temporal leakage. train=800 test=200 cutoff=2026-02-18 11:00:00
2	xray*to*result*min is NaN exactly when has*xray == 0	PASS	X-ray gaps are exactly the no-X-ray cases
3	lab*turnaround*min is NaN exactly when has_lab == 0	PASS	Lab gaps are exactly the no-lab cases
4	loopback*delay*min is NaN exactly when has_reassessment == 0	PASS	Rework delay gaps are exactly the no-rework cases
5	genuine missing values in observed columns	PASS	Documents that nothing needed imputing. found=0
6	no missing values remain after imputation	PASS	Final tables are complete
7	imputation benchmark completed	PASS	Records which method won. best by mean NRMSE: iterative*bayesridge; iterative*bayesridge=0.61...
8	standardized train: mean ~ 0	PASS	Training mean is zero after standardising. max|mean|=1.27e-16
9	standardized train: std ~ 1	PASS	Training std is one
10	normalized train within [0,1] and hits both ends	PASS	Training range is [0,1] with both ends reached
11	normalized test values outside [0,1] (expected possible, NOT clipped)	PASS	Test extrapolation is visible, not hidden. count=2
12	inverse transform recovers original values	PASS	Scaling is invertible (no information lost)

#	Check	Result	Purpose / detail
13	event-level scaled rows == events rows	PASS	No event row lost in scaling. 6964
14	every event case has a split	PASS	Every event maps to a split
15	model-ready features have no NaN	PASS	Features are complete
16	one-hot/sin/cos columns untouched (already bounded)	PASS	Encoded bounded columns left alone

Table 7.4 Checks recorded in scaling_checks.csv

7.9 Design decisions

#	Decision	Reason	Alternative rejected
1	Cut the data by time, not at random	Predictions will be made on future patients	Random 80/20 split
2	Zero-fill structural gaps and add flags	Mean-fill would fabricate procedures	Mean, median or model-based imputation of absent steps
3	Benchmark imputers even though none was needed	Evidence for future data; protects against arbitrary choice	Assume mean imputation
4	Fit scalers on training rows only	Test statistics must stay unseen	Fit on the whole file
5	Do not clip Min-Max test values	Preserves evidence of extrapolation	Clip to [0, 1]
6	Keep raw and scaled columns in the event file	Analysts can read minutes while models use scaled values	Scaled columns only
7	Feature store limited to arrival-time fields	Prevents leakage	Include interval and event-count columns
8	Save scaler parameters to JSON	Live admissions must be transformed identically	Refit at inference

7.10 Limitations

- The upstream cleaning script and its two input files are not in the reviewed branches, so the module cannot be run end to end from the repository alone.
- Failed checks do not stop execution; a reviewer must read the PASS/FAIL summary.
- The imputation benchmark runs on a single random mask (seed 42); averaging over several masks would give a confidence range.
- Physician identifiers are one-hot encoded, which fixes the physician set at 12; a new physician needs a re-run.
- The `main` README refers to a "15-point" log while the shipped log has 16 checks.

8 Process-Mining Analysis Modules

This chapter specifies the 18 Python scripts in `prachi/src`. They share a common design, described first, and are then documented one by one in the form of module cards.

8.1 Common design pattern

Sixteen of the 18 scripts read the full event log `careflow_events_scaled.csv`; the other two are early prototypes that read the three-case sample. All follow the same five-stage pattern.

Stage	Name	Behaviour
1	Locate	Resolve the project root as the parent of `src/` and build input and output paths from it.
2	Validate	Check that required columns exist, that none contains nulls, and that timestamps parse. Any failure raises `ValueError`.
3	Order	Sort by `case_id` and `activity_order` so each case reads as a sequence.
4	Derive	Compute the module-specific measure (sequences, shifted next-activity columns, elapsed minutes, counts).
5	Export	Write a summary CSV into `data/processed/`, print a console summary, and (for documentation) save a markdown write-up in `docs/`.

Core derivation: elapsed time between consecutive events

The transition and elapsed-time modules (M-03, M-06, M-10 to M-13) rely on one idiom: shifting the activity and timestamp columns by one row within each case.

Elapsed-time derivation (from process_bottleneck_analysis.py)

```
df = df.sort_values(["case_id", "activity_order"]).copy()
df["next_activity"]  = df.groupby("case_id")["activity_name"].shift(-1)
df["next_timestamp"] = df.groupby("case_id")["timestamp"].shift(-1)
df["elapsed_minutes"] = (df["next_timestamp"] - df["timestamp"]).dt.total_seconds() / 60
transitions = df.dropna(subset=["next_activity", "next_timestamp"])
transitions = transitions[transitions["elapsed_minutes"] >= 0]
```

Because the last event of each case has no successor, `dropna` removes exactly 1,000 rows, leaving 5,964 transitions. Negative intervals are discarded as invalid; none occur in the data.

8.2 Module index

ID	Script	Question answered	Output
M-01	`validate_event_log.py`	Is the log well formed?	Console report
M-02	`loopback_analysis.py`	How often are steps repeated?	4 loopback/repeat CSVs
M-03	`activity_transition_analysis.py`	Which steps follow which?	Transition CSVs
M-04	`patient_workflow_discovery.py`	What does the process look like?	`patient_workflow_map.svg`
M-05	`pathway_variation_analysis.py`	Which pathways exist?	`pathway_variation_summary.csv`
M-06	`patient_waiting_time_analysis.py`	How long between steps?	`patient_waiting_time_summary.csv`
M-07	`activity_frequency_analysis.py`	How often is each activity recorded?	`activity_frequency_summary.csv`
M-08	`activity_distribution_analysis.py`	How many patients reach each activity?	`activity_distribution_summary.csv`

ID	Script	Question answered	Output
M-09	`patient_journey_sequence_analysis.py`	What are the complete journeys?	`patient_journey_sequence_summary.csv`
M-10	`patient_journey_transition_analysis.py`	How is traffic split across transitions?	`patient_journey_transition_summary.csv`
M-11	`process_bottleneck_analysis.py`	Which transitions take longest?	`process_bottleneck_summary.csv`
M-12	`workflow_performance_summary.py`	How do frequency and time combine?	`workflow_performance_summary.csv`
M-13	`workflow_improvement_recommendations.py`	What should be reviewed first?	`workflow_improvement_recommendations.csv`
M-14	`workflow_visualization_dashboard.py`	Can results be charted?	3 PNG charts + details CSV
M-15	`careflow_interactive_dashboard.py`	Can managers explore results?	Streamlit app
M-16	`patient_journey_comparison.py`	How do pathways compare in duration?	`patient_journey_comparison_summary.csv`, report
M-17	`process_mining.py`	Prototype process map on sample log	`process_map.svg`
M-18	`waiting_time_analysis.py`	Prototype elapsed time on sample log	Sample CSVs

8.3 Module specifications

M-01	validate_event_log.py: event-log validation
Purpose	Confirm that an event log is fit for process mining before any analysis.
Input	`data/raw/sample_event_log.csv` (16 events, 3 cases; columns `case_id`, `activity`, `timestamp`).
Logic	Five checks: (1) required columns present; (2) no missing values in them; (3) all timestamps parse; (4) events can be arranged chronologically within each case (no negative gap after sorting); (5) count exact duplicate rows (warning only).
Output	Console report of PASS / FAIL / WARNING lines.
Result	All five checks pass on the sample log.
Limitation	Runs only on the prototype sample. The full log is validated implicitly by the guard clauses of every later script, but duplicates are not re-checked.

M-02	loopback_analysis.py: repeated activities and loopbacks
Purpose	Quantify rework: events that repeat an earlier activity or belong to a loop back to an earlier stage.
Input	`careflow_events_scaled.csv`; uses `is_repeat_activity`, `is_loopback_event`, `activity_occurrence` and related columns.
Logic	Count events flagged as loopback and repeated, group by activity, and identify the number of cases affected and the sequences involved.
Output	Four CSVs in `data/processed/` covering repeat-activity and loopback summaries.
Result	705 loopback events across 235 cases (23.5%): 235 each of repeat Triage, Doctor Reassessment and Documentation Review. The only repeated activity is Triage (second occurrence).
Limitation	Reason codes are read as recorded; every one is "Missing intake paperwork".

M-03	activity_transition_analysis.py: consecutive-activity transitions
Purpose	Count how often one activity is directly followed by another.
Input	`careflow_events_scaled.csv` (`case_id`, `activity_name`, `activity_order`).
Logic	Shift the activity column within each case, count pairs, compute each pair's share of all transitions.
Output	Transition count tables in `data/processed/`.
Result	16 distinct transition types, 5,964 transitions in total (Figure 8.1).

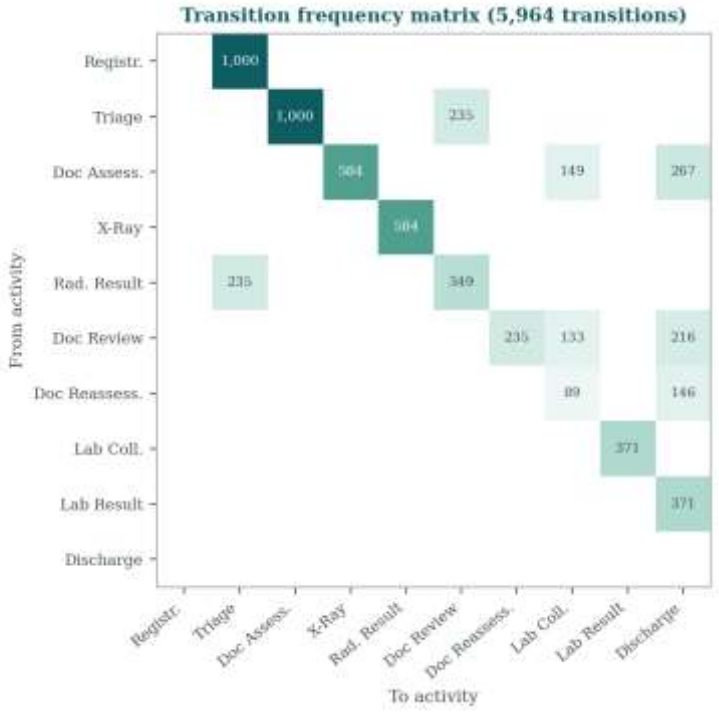


Figure 8.1 Transition frequency matrix. Rows are the source activity, columns the target

M-04	patient_workflow_discovery.py: process discovery
Purpose	Discover and draw the directly-follows graph of the process.
Input	`careflow_events_scaled.csv` (`case_id`, `activity_name`, `timestamp`, `activity_order`).
Logic	Rename columns to the PM4Py standard (`case:concept:name`, `concept:name`, `time:timestamp`), convert to an event log, discover the DFG with frequencies, and render it as SVG.
Output	`docs/patient_workflow_map.svg`.
Result	All 1,000 cases start at Registration and end at Discharge. The dominant trunk is Registration, Triage, Doctor Assessment. Figure 8.2 shows the same graph redrawn in the project theme, with transition count and mean elapsed minutes on every edge.

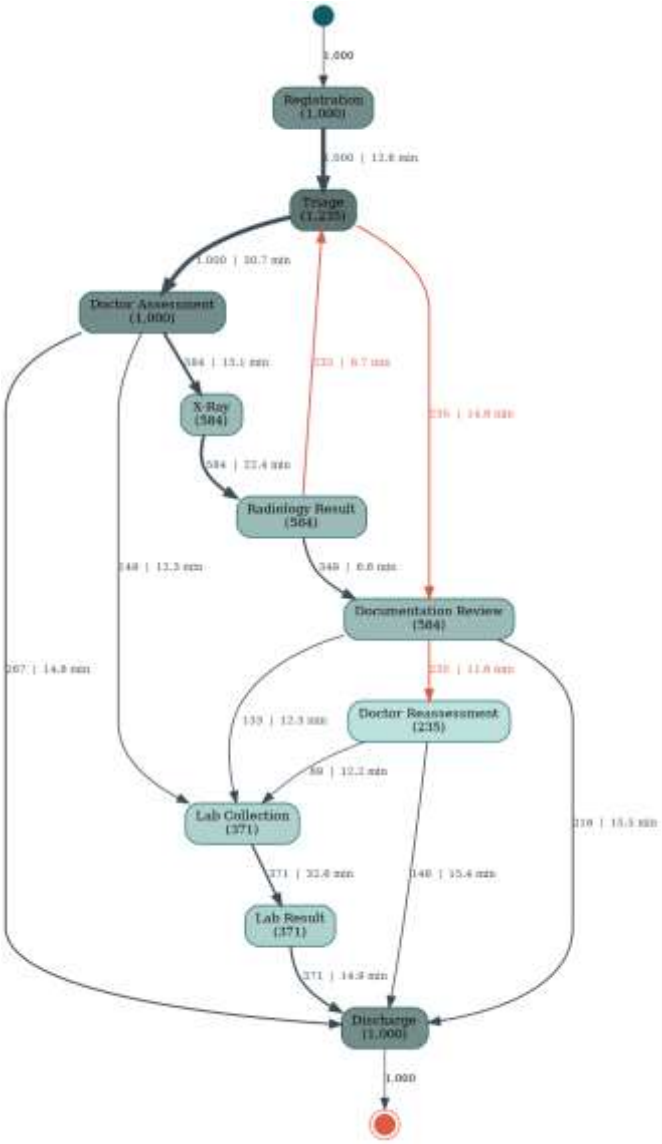


Figure 8.2 Directly-follows graph. Edge labels show count and mean elapsed minutes; coral edges form the rework loop

M-05	pathway_variation_analysis.py: pathway variants
Purpose	Find every distinct complete pathway and its share.
Input	`careflow_events_scaled.csv`.
Logic	Join activity names in order for each case with " -> ", count identical sequences, compute percentage of cases.
Output	`pathway_variation_summary.csv`.
Result	Six variants. The leading pathway (Registration, Triage, Doctor Assessment, Discharge) covers 267 cases (26.7%); the two rework variants together cover 235 (23.5%).

ID	Cases	Share	Pathway
V1	267	26.7%	Registration → Triage → Doctor Assessment → Discharge
V2	216	21.6%	Registration → Triage → Doctor Assessment → X-Ray → Radiology Result → Documentation Review → Discharge
V3	149	14.9%	Registration → Triage → Doctor Assessment → Lab Collection → Lab Result → Discharge
V4	146	14.6%	Registration → Triage → Doctor Assessment → X-Ray → Radiology Result → Triage → Documentation Review → Doctor Reassessment → Discharge
V5	133	13.3%	Registration → Triage → Doctor Assessment → X-Ray → Radiology Result → Documentation Review → Lab Collection → Lab Result → Discharge
V6	89	8.9%	Registration → Triage → Doctor Assessment → X-Ray → Radiology Result → Triage → Documentation Review → Doctor Reassessment → Lab Collection → Lab Result → Discharge

Table 8.1 The six pathway variants (M-05)

M-06	patient_waiting_time_analysis.py: elapsed time per transition
Purpose	Measure the minutes between consecutive events for every transition type.
Input	`careflow_events_scaled.csv`; requires `case_id`, `activity_name`, `activity_order`, `timestamp`.
Logic	Elapsed-time derivation (Section 8.1); group by (from, to); compute count, mean, median and maximum; round to two decimals.
Output	`patient_waiting_time_summary.csv`.
Result	The slowest transitions are Lab Collection to Lab Result (32.6 min), Triage to Doctor Assessment (30.7) and X-Ray to Radiology Result (22.4). Table 8.2 gives all 16 with percentiles computed for this document.
Limitation	File name says "waiting time" but the measure is elapsed time, which includes activity duration.

Transition	n	Mean	Median	P90	P95	Max	Std
Lab Collection to Lab Result	371	32.6	32.0	48.0	49.0	50.0	10.4
Triage to Doctor Assessment	1,000	30.7	29.0	55.0	58.0	60.0	16.3
X-Ray to Radiology Result	584	22.4	22.0	33.0	34.0	35.0	7.5
Documentation Review to Discharge	216	15.5	16.0	23.0	24.0	25.0	5.9
Doctor Reassessment to Discharge	146	15.4	16.0	24.0	24.8	25.0	6.1
Doctor Assessment to X-Ray	584	15.1	15.0	24.0	24.0	25.0	6.0
Lab Result to Discharge	371	14.9	14.0	23.0	24.0	25.0	6.1
Triage to Documentation Review	235	14.8	15.0	24.0	24.3	25.0	6.1
Doctor Assessment to Discharge	267	14.8	15.0	23.0	24.0	25.0	6.3
Registration to Triage	1,000	12.6	12.0	22.0	24.0	25.0	6.8
Doctor Assessment to Lab Collection	149	12.3	13.0	19.0	20.0	20.0	4.7
Documentation Review to Lab Collection	133	12.3	12.0	18.0	19.4	20.0	4.5
Doctor Reassessment to Lab Collection	89	12.2	13.0	19.0	20.0	20.0	4.5
Documentation Review to Doctor Reassessment	235	11.6	12.0	18.0	20.0	20.0	4.9
Radiology Result to Triage	235	8.7	8.0	14.0	15.0	15.0	4.0
Radiology Result to Documentation Review	349	6.6	7.0	11.0	11.0	11.0	2.7

Table 8.2 Elapsed minutes by transition. P90 and P95 recomputed from the event log

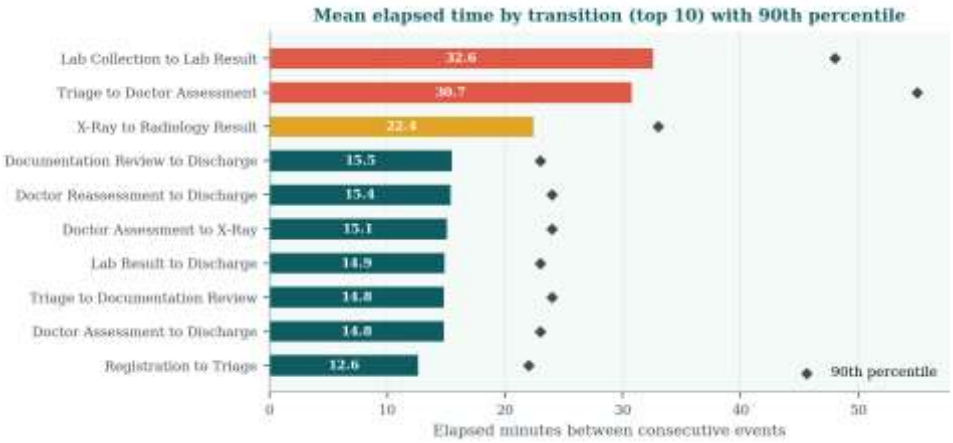


Figure 8.3 Ten transitions with the highest mean elapsed time

M-07	activity_frequency_analysis.py: activity frequency
Purpose	Count events per activity, share of total events, and repeated events.
Input	`careflow_events_scaled.csv` (`case_id`, `activity_name`, `activity_order`).
Output	`activity_frequency_summary.csv` with `event_count`, `unique_cases`, `percentage_of_events`, `percentage_of_cases`, `repeat_event_count`.
Result	Triage 1,235 events (17.7%), of which 235 are repeats; Registration, Doctor Assessment and Discharge 1,000 each (14.4%); imaging activities 584 (8.4%); laboratory 371 (5.3%); Doctor Reassessment 235 (3.4%).

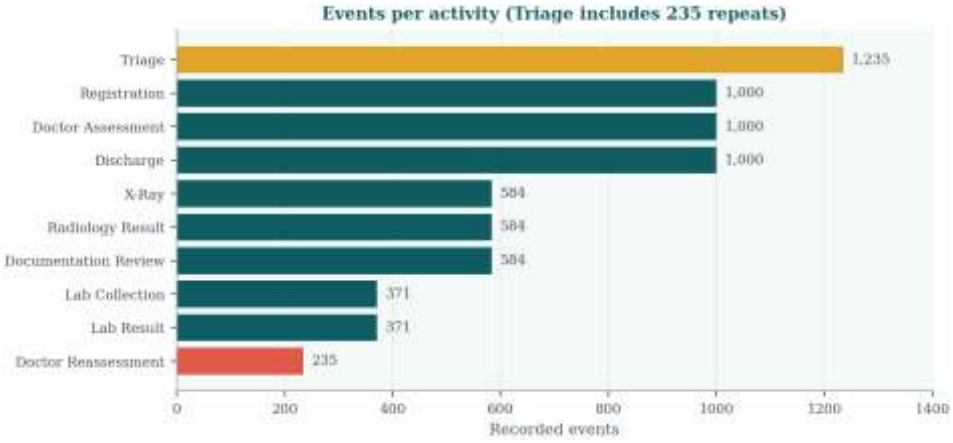


Figure 8.4 Events per activity

M-08	activity_distribution_analysis.py: patients per activity
Purpose	Separate "how many events" from "how many patients" for each activity.
Output	`activity_distribution_summary.csv` with `event_count`, `unique_patients`, `percentage_of_events`, `percentage_of_patients`.
Result	Three activities reach every patient; X-ray work reaches 58.4%, laboratory 37.1%, reassessment 23.5%. Triage has 1,235 events but only 1,000 patients, which exposes repeat triage.
Limitation	Says nothing about speed or efficiency.

M-09	patient_journey_sequence_analysis.py: complete journeys
Purpose	Rank complete activity sequences by the number of patients following them.
Output	`patient_journey_sequence_summary.csv` with `rank`, `activity_sequence`, `patient_count`, `activity_count`, `percentage_of_patients`. This file feeds the dashboard and chart script.
Result	Six journeys, from 4 activities (fast track, 267 patients) to 10 activities (rework plus laboratory, 89 patients).

M-10	patient_journey_transition_analysis.py: transition share
Purpose	Express each transition type as a share of all 5,964 transitions.
Output	`patient_journey_transition_summary.csv` with `rank`, `from_activity`, `to_activity`, `transition_count`, `percentage_of_transitions`.
Result	Table 8.3. The two intake transitions (Registration to Triage, Triage to Doctor Assessment) each carry 16.8% of all traffic.

Rank	Transition	Count	% of transitions	Cumulative %
1	Registration to Triage	1,000	16.77	16.8
2	Triage to Doctor Assessment	1,000	16.77	33.5
3	Doctor Assessment to X-Ray	584	9.79	43.3
4	X-Ray to Radiology Result	584	9.79	53.1
5	Lab Collection to Lab Result	371	6.22	59.3
6	Lab Result to Discharge	371	6.22	65.6
7	Radiology Result to Documentation Review	349	5.85	71.4
8	Doctor Assessment to Discharge	267	4.48	75.9
9	Documentation Review to Doctor Reassessment	235	3.94	79.8
10	Radiology Result to Triage	235	3.94	83.8
11	Triage to Documentation Review	235	3.94	87.7
12	Documentation Review to Discharge	216	3.62	91.3
13	Doctor Assessment to Lab Collection	149	2.50	93.8
14	Doctor Reassessment to Discharge	146	2.45	96.3
15	Documentation Review to Lab Collection	133	2.23	98.5
16	Doctor Reassessment to Lab Collection	89	1.49	100.0

Table 8.3 Transition share (M-10). Order among ties follows alphabetical order

M-11	process_bottleneck_analysis.py: bottleneck candidates
Purpose	Rank transitions by mean elapsed time so analysts know where to look first.
Logic	Same derivation as M-06; sort by mean elapsed time, descending; insert rank.
Output	`process_bottleneck_summary.csv`.
Result	Ranks 1 to 3 are Lab Collection to Lab Result, Triage to Doctor Assessment and X-Ray to Radiology Result. All three involve a diagnostic result or the wait for the doctor.
Interpretation	A candidate is a transition worth investigating. The module deliberately does not claim causes.

M-12	workflow_performance_summary.py: combined frequency and time
Purpose	Provide one table that joins how often a transition occurs with how long it takes.
Output	`workflow_performance_summary.csv` (columns include transition counts, average, median and maximum elapsed minutes). Used by the dashboard and chart script.
Result	Same figures as Table 8.2 for the 16 transitions; adds a ranked view.

M-13	workflow_improvement_recommendations.py: review recommendations
Purpose	Convert the elapsed-time table into a prioritised list of things to review, each with a suggested metric.
Logic	Six transitions have hand-written recommendations and are labelled "High investigation priority". Other transitions with mean elapsed time of 15 minutes or more are "Further review"; the rest "Routine monitoring".
Output	`workflow_improvement_recommendations.csv` (16 rows, columns `recommendation_id`, `from_activity`, `to_activity`, `transition_count`, `average_elapsed_minutes`, `median_elapsed_minutes`, `investigation_priority`, `recommendation`, `suggested_metric`).
Result	6 High, 1 Further review, 9 Routine. The sorted output places "Further review" first because the priority label is sorted alphabetically (Section 15, issue I-09).
Design note	The script prints a reminder that elapsed time alone does not prove a bottleneck or its cause.

#	Transition	Mean	Priority	Suggested focus
1	Lab Collection to Lab Result	32.6	High	Sample transport, processing queue, result time-stamping
2	Triage to Doctor Assessment	30.7	High	Queue pattern by hour and acuity before changing staffing
3	X-Ray to Radiology Result	22.4	High	Imaging queue, reporting workflow, timestamp consistency
4	Documentation Review to Discharge	15.5	High	Documentation completeness, discharge approval
5	Doctor Reassessment to Discharge	15.4	High	Reassessment documentation and following steps
6	Radiology Result to Triage	8.7	High	Why triage occurs again (rework path)
7	Doctor Assessment to X-Ray	15.1	Further review	Validate timestamps, sample cases
8-16	Remaining nine transitions	6.6 to 14.9	Routine	Monitor count and median

Table 8.4 Recommendation summary from M-13

M-14	workflow_visualization_dashboard.py: static charts
Purpose	Create three publication-quality PNG charts from the processed summaries.
Inputs	`activity_frequency_summary.csv`, `patient_journey_sequence_summary.csv`, `workflow_performance_summary.csv`. Aborts with `FileNotFoundError` if any is missing.
Logic	A helper `find_column(df, candidates)` locates a column by several possible names (case-insensitive) so the script tolerates naming variations.
Output	`docs/activity_frequency_chart.png`, `docs/patient_pathway_chart.png`, `docs/workflow_elapsed_time_chart.png`, and `workflow_visualization_pathway_details.csv` (maps chart labels to full sequences).

M-15	careflow_interactive_dashboard.py: Streamlit dashboard
Purpose	Let managers explore activity frequency, pathways and elapsed-time transitions.
Inputs	The same three summary CSVs as M-14, read from `data/processed/`.
Interface	Wide-layout page titled "CareFlow Patient Workflow Dashboard"; four KPI cards; three numbered sections with bar charts and data tables (Chapter 11).
Run	`streamlit run src/careflow_interactive_dashboard.py`.
Resilience	Uses `find_column` to tolerate renamed columns; displays an error message when files are missing.

M-16	patient_journey_comparison.py: pathway duration comparison
Purpose	Compare complete pathways by frequency and by journey duration.
Input	`careflow_events_scaled.csv`.
Logic	Build an activity sequence and the first-to-last timestamp duration for each case; group by sequence; compute patient count, mean, minimum, maximum and percentage.
Output	`patient_journey_comparison_summary.csv` and a markdown report `docs/patient_journey_comparison.md` written by the script itself.
Result	Table 8.5 (agrees with the case-level figures in Section 10.3).

ID	Pathway	Cases	Mean (min)	Min	Max	Share
V1	Reg → Triage → Assess → Discharge	267	58.5	12	107	26.7%
V2	Reg → Triage → Assess → X-Ray → Radiology Result → Documentation Review → Discharge	216	102.3	52	158	21.6%
V3	Reg → Triage → Assess → Lab Collection → Lab Result → Discharge	149	102.9	44	157	14.9%
V4	Reg → Triage → Assess → X-Ray → Radiology Result → Triage → Documentation Review → Doctor Reassessment → Discharge	146	132.8	73	192	14.6%
V5	Reg → Triage → Assess → X-Ray → Radiology Result → Documentation Review → Lab Collection → Lab Result → Discharge	133	146.6	84	207	13.3%
V6	Reg → Triage → Assess → X-Ray → Radiology Result → Triage → Documentation Review → Doctor Reassessment → Lab Collection → Lab Result → Discharge	89	174.6	120	234	8.9%

Table 8.5 Pathway comparison (M-16). Common prefix abbreviated

M-17	process_mining.py: prototype process map
Purpose	First PM4Py experiment on the 3-case sample log (Day 3 era).
Input / output	`sample_event_log.csv` to `docs/process_map.svg`.
Status	Superseded by M-04. Retained as a historical prototype; its activities (for example "Doctor Examination", "Treatment") differ from the final data set.

M-18	waiting_time_analysis.py: prototype elapsed-time analysis
Purpose	First elapsed-time experiment on the sample log.
Output	`activity_elapsed_times.csv`, `transition_waiting_summary.csv`, `patient_journey_summary.csv`.
Status	Superseded by M-06 and M-16. Its CSVs describe only the 3-case sample and must not be mixed with full-data results.

8.4 Dependencies between modules

Analysis modules M-02 to M-13 and M-16 each read the event log directly and are independent of one another, so they can be run in any order. M-14 and M-15 depend on the outputs of M-07, M-09 and M-12. The documents in `docs/` are written by hand from the console output (except M-16, which writes its own report).

Consumer	Requires output of	Failure if missing
M-14 charts	M-07, M-09, M-12	`FileNotFoundError` naming the file
M-15 dashboard	M-07, M-09, M-12	Dashboard shows an error message
All other M-modules	Event log only	`FileNotFoundError` from `pd.read_csv`

8.5 Module-level limitations

- The 16 full-log scripts repeat the same load, validate and sort code. A shared helper module would remove about 20 lines of duplication per script and reduce the risk of the copies drifting apart.
- Write-ups are hand-authored and can drift from the CSVs; scripts have no unit tests, so verification relies on cross-checking totals between modules (6,964 events, 5,964 transitions, 1,000 cases).
- Elapsed time is reported for transitions only; no module reports queue length or resource utilisation, which would need resource-level timestamps.

9 Exploratory Analysis Notebooks

This chapter documents the 14 Jupyter notebooks in `sagar/notebooks`. They complement the Python modules: where modules produce a fixed table, notebooks explore, chart and audit. Notebook IDs N-03 to N-16 follow the file name day number.

9.1 Common notebook design

- Each notebook loads data from `data/processed_new/` (cases file for case-level work, `careflow_events_scaled.csv` for event-level work) using a relative path.
- Analysis cells are followed by a markdown cell titled "Key Insights" that states the finding in plain language.
- Notebooks from N-10 onward also save a summary artefact (CSV) that later notebooks reload, so they must be run in order.

9.2 Notebook index

ID	Notebook	Input	Output / artefact
N-03	`day3_eda_baseline`	Cases (1000 x 32)	Schema and data-type audit
N-04	`day4_demographics_analysis`	Cases	Age, acuity and department charts
N-05	`day5_activity_frequency`	Events (6964 x 27)	Top-10 activity chart
N-06	`day6_process_timeline`	Cases	LOS statistics and boxplot
N-07	`day7_trace_sequence_mining`	Events	Traces, variant counts, loopback cases
N-08	`day8_department_transition`	Events	Transition matrix and heatmap
N-09	`day9_acuity_wait_time_correlation`	Cases	Correlation matrix and boxplot
N-10	`day10_mid_project_performance_audit`	Cases, events	`day10_audit_summary.csv`
N-11	`day11_outlier_and_anomaly_detection`	Cases	`careflow_cases_outliers.csv`
N-12	`day12_cohort_segmentation`	Outlier file	`careflow_cases_segmented.csv`
N-13	`day13_time_series_peak_loads`	Events	`day13_peak_loads_summary.csv`
N-14	`day14_feature_significance`	Cases	`day14_feature_significance_summary.csv`
N-15	`day15_manager_insights_dashboard`	N-13 and N-14 outputs	`day15_executive_dashboard_summary.csv`
N-16	`day16_final_metrics_summary`	Cases, N-10	`careflow_final_metrics_summary.csv`

9.3 Notebook specifications

ID	Purpose and method	Observed result	Note
N-03	Baseline EDA and schema audit. Confirm the master case table is loaded correctly and that imputation left no gaps. Loads the imputed cases file, prints shape (1000, 32), head and `info()`.	Shape (1000, 32); all columns non-null; types as expected (Section 6.3).	-
N-04	Demographics. Describe who the patients are. Count plots for age group, acuity and department; histogram of age.	Age 18 to 90 (mean 55.0); acuity peaks at ESI 3; one department.	The department chart has a single bar.
N-05	Activity frequency. Rank activities by event count. Counts `activity_name`, plots the top 10.	Triage 1,235; Registration, Doctor Assessment, Discharge 1,000; X-ray group 584; lab group 371; reassessment 235.	Matches M-07.
N-06	Process timeline and LOS. Describe end-to-end duration. Summary statistics of `total_duration_min`; boxplot by department.	Mean 107.50, SD 43.52, min 12, Q1 76, median 105, Q3 138.25, max 234.	Department boxplot is trivial (one department).

ID	Purpose and method	Observed result	Note
N-07	Trace and sequence mining. Build traces and count variants; detect loopbacks. Groups events by case into `trace_sequence`, counts frequency and percentage, flags cases with repeated activities.	Top pathways 267 (26.7%), 216 (21.6%), 149 (14.9%); 235 cases (23.5%) with loopbacks.	Matches M-05 and M-02.
N-08	Transition matrix. Show which activity follows which. Cross-tabulates `from_activity` by `to_activity`; heatmap.	For example Doctor Assessment to Discharge 267, Doctor Reassessment to Discharge 146.	Counts only; no timing.
N-09	Acuity versus duration. Test whether urgency relates to stay length. Merges acuity, age and duration; Pearson correlation; boxplot by acuity.	Acuity vs duration r = 0.232; age vs duration r = -0.042; acuity vs age r = 0.023.	The column named `total_wait_time_min` is a copy of `total_duration_min` (correlation 1.0), so it is not a separate wait measure.
N-10	Mid-project audit. Record a baseline of headline metrics. Builds a one-table summary: total cases, total events, average "wait time", critical cases, departments.	1,000 cases; 6,964 events; average 64.135; 170 critical cases; 1 department.	"Avg Wait Time 64.135" is the mean assessment-to-discharge interval; the markdown says about 107.5 (total duration). See issue I-03.
N-11	Outlier and anomaly detection. Flag unusually long stays. IQR rule (Q1 - 1.5 IQR, Q3 + 1.5 IQR) on `assessment_to_discharge_min`; saves tagged file.	Q1 24.00, Q3 93.25, IQR 69.25, upper threshold 197.12 min; 0 outliers.	Max value is 176, below the threshold. See issue I-01.
N-12	Cohort segmentation. Compare critical and standard patients. Splits into High Acuity / Critical (ESI 1-2) and Standard Acuity; summarises assessment-to-discharge by cohort and outlier flag.	Critical: 170 cases, mean 62.44, median 61, max 176. Standard: 830 cases, mean 64.48, median 64, max 172.	Near-identical times suggest the data does not give critical patients faster downstream care.
N-13	Time series and peak loads. Find busy hours. Counts events per hour of day; line chart; saves summary.	Event counts per hour range from about 250 to about 350, peaking at 16:00 to 18:00.	Column is named `patient_count` but counts events. The saved summary labels every hour "Normal" (placeholder). See issue I-07.
N-14	Feature significance. Rank drivers of duration. Fits `RandomForestRegressor(random_state=42)` and plots importances.	Only one feature (acuity) is used; importance 1.0.	With a single feature the ranking is uninformative. See issue I-07.
N-15	Manager insights dashboard. Consolidate N-13 and N-14. Loads both artefacts and draws a multi-panel view; saves `day15_executive_dashboard_summary.csv`.	Inherits the content of N-13 and N-14.	Only as good as its inputs.
N-16	Final metrics summary. Gather the headline metrics. Computes cases, mean LOS (107.50), mean door-to-doctor (43.40); builds a bottleneck table.	Table lists four bottleneck pairs with delays 32.6, 30.7, 22.4 and 15.1.	Outlier count (80) and bottleneck names are typed in. See issues I-01 and I-02.

Table 9.0 Notebook specifications

9.4 Cohort comparison from N-12

Cohort	Cases	Mean (min)	Median	Max	Outliers
High Acuity / Critical (ESI 1-2)	170	62.44	61.0	176.0	0
Standard Acuity (ESI 3-5)	830	64.48	64.0	172.0	0

Table 9.1 Assessment-to-discharge time by cohort (N-12)

Critical patients reach the doctor much faster (about 15 minutes against 49, Section 10.5), yet their time to discharge is almost the same as that of standard patients. In a real department sicker patients would normally take longer; here the similarity is a property of the synthetic data.

10 Analysis Results

This chapter documents what the modules and notebooks produce when run on the supplied data. Figures were recomputed directly from `careflow_cases_imputed.csv` and `careflow_events_scaled.csv`, so they are consistent with each other. Where a figure differs from a number published in a branch README, Chapter 15 records the difference.

10.1 Activity profile

Ten activities account for 6,964 events. Four activities (Registration, Triage, Doctor Assessment, Discharge) form the compulsory spine every patient passes through; the other six are conditional on what the doctor orders. Triage occurs more often than patients because 235 patients are triaged a second time.

Activity	Events	Cases	% of cases	% of events	Role
Triage	1,235	1,000	100.0	17.73	Spine (235 repeats)
Registration	1,000	1,000	100.0	14.36	Spine
Doctor Assessment	1,000	1,000	100.0	14.36	Spine
Discharge	1,000	1,000	100.0	14.36	Spine
X-Ray	584	584	58.4	8.39	Conditional: imaging
Radiology Result	584	584	58.4	8.39	Conditional: imaging
Documentation Review	584	584	58.4	8.39	After imaging
Lab Collection	371	371	37.1	5.33	Conditional: laboratory
Lab Result	371	371	37.1	5.33	Conditional: laboratory
Doctor Reassessment	235	235	23.5	3.37	Rework only

Table 10.1 Activity profile

Diagnostic use is easy to summarise: 267 cases (26.7%) had neither X-ray nor laboratory work, 362 had X-ray only, 149 laboratory only and 222 both. Together 733 patients (73.3%) used at least one diagnostic service.

10.2 Pathway variants

Section 6.1 of the data design lists six variants. Their durations form three groups: the fast track (V1) is about an hour; single-diagnostic pathways (V2, V3) take about 102 minutes; pathways with rework or two diagnostics take 133 minutes or longer.

ID	Type	Cases	Mean	Median	SD	Range
V1	Fast track	267	58.5	58.0	21.2	12-107
V2	Radiology	216	102.3	102.0	22.1	52-158
V3	Laboratory	149	102.9	102.0	23.7	44-157
V4	Radiology + rework	146	132.8	134.0	24.2	73-192
V5	Radiology + laboratory	133	146.6	147.0	26.2	84-207
V6	Radiology + rework + lab	89	174.6	175.0	26.8	120-234

Table 10.2 Duration statistics by pathway variant (minutes)

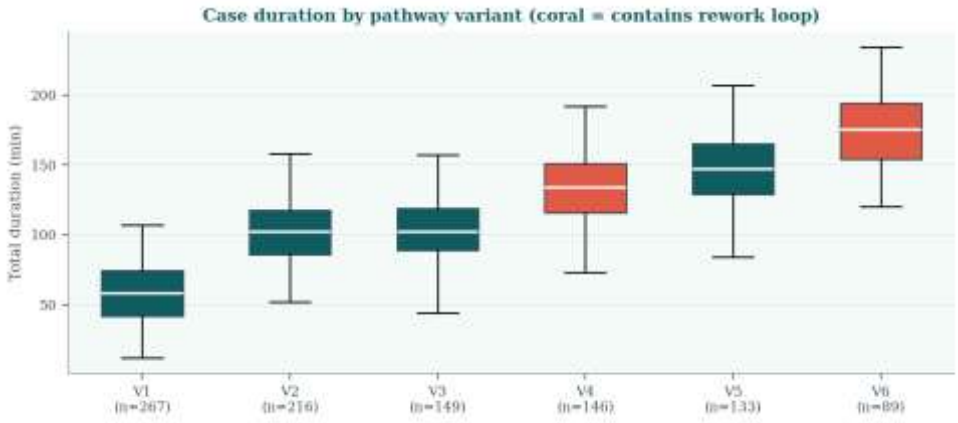


Figure 10.1 Case duration by variant. Coral boxes contain the rework loop

Where the time goes

Splitting each journey into six stages shows which part of the process makes a pathway long. Intake (about 12.5 min) and the wait for a doctor (about 30 min) are nearly constant in every variant, so they cannot explain differences between variants. Differences come from diagnostics (imaging about 37 min, laboratory about 44 min) and, in V4 and V6, from rework (about 35 min).

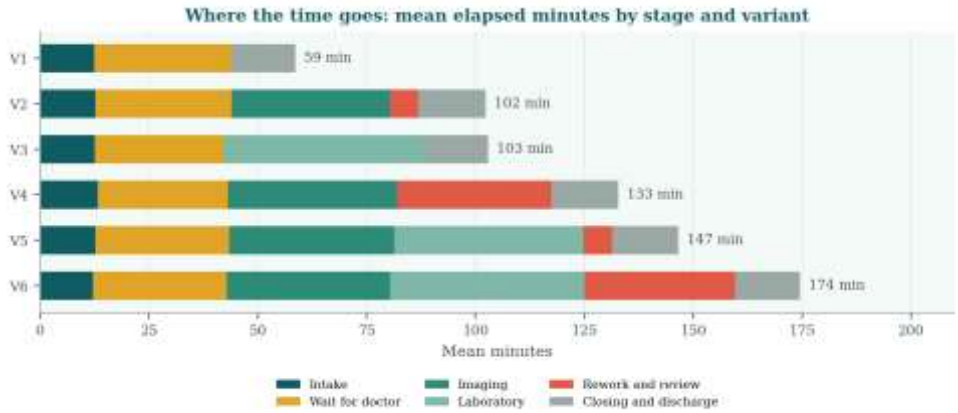


Figure 10.2 Mean minutes per stage in each variant

ID	Intake	Wait for doctor	Imaging	Laboratory	Rework and review	Closing and discharge	Total
V1	12.4	31.4	0.0	0.0	0.0	14.8	58.6
V2	12.7	31.3	36.3	0.0	6.5	15.5	102.3
V3	12.5	29.7	0.0	46.1	0.0	14.6	102.9
V4	13.3	29.8	38.9	0.0	35.4	15.4	132.8
V5	12.7	30.8	37.8	43.5	6.6	15.2	146.6
V6	12.2	30.6	37.6	44.6	34.6	14.9	174.5

Table 10.3 Mean minutes by stage and variant. Stage definitions in the notes below

Stage definitions: Intake is Registration to Triage; Wait for doctor is Triage to Doctor Assessment; Imaging is Doctor Assessment to X-Ray plus X-Ray to Radiology Result; Laboratory is every transition leading into or out of Lab Collection; Rework and review covers Radiology Result to Triage, Triage to Documentation Review, Documentation Review to Doctor Reassessment and Radiology Result to Documentation Review; Closing is everything that ends in Discharge.

10.3 Case duration distribution

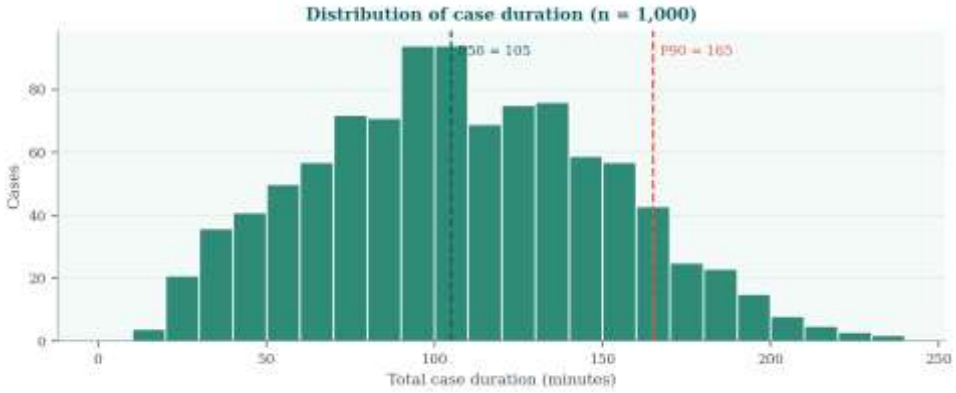


Figure 10.3 Histogram of total case duration

Measure (min)	P5	P10	P25	P50	P75	P90	P95	P99
Total duration	37.0	49.0	76.0	105.0	138.2	165.0	181.0	208.0
Registration to triage	2.0	4.0	7.0	12.0	19.0	22.0	24.0	25.0
Door to doctor	12.0	17.0	28.0	44.0	59.0	69.0	74.0	80.0
Assessment to discharge	7.0	12.0	24.0	64.0	93.2	118.0	132.0	153.0
X-ray turnaround*	11.0	12.0	16.0	22.0	29.0	33.0	34.0	35.0
Lab turnaround*	17.0	18.0	24.0	32.0	41.5	48.0	49.0	50.0
Loopback delay*	19.0	23.0	29.0	35.0	41.0	47.0	50.3	55.0

*Table 10.4 Percentiles of key measures. *Among cases where the step occurred*

The distribution is wide (SD 43.5 minutes) and mildly right-skewed (skewness 0.189). One patient in ten stays longer than 165 minutes and one in twenty longer than 181. Reporting only the mean (107.5) would hide this spread, which is why Chapter 17 recommends tracking P50 and P90.

Duration band (min)	Cases	Share	Comment
0 to 30	26	2.6%	Very short fast-track visits
30 to 60	131	13.1%	Mostly fast track
60 to 90	204	20.4%	Core of single-diagnostic pathways
90 to 120	259	25.9%	Around the median
120 to 150	202	20.2%	Longer pathways
150 to 180	126	12.6%	Two diagnostics or rework
180 to 240	52	5.2%	Longest stays

Table 10.5 Cases by duration band

10.4 Rework (loopbacks)

Rework is the most structured pattern in the data. Every one of the 235 rework cases follows the same loop: after the Radiology Result the patient returns to Triage, then passes through Documentation Review and Doctor Reassessment before continuing. The loop exists only in pathways that include imaging.

Measure	Value	Notes
Cases with rework	235 (23.5%)	V4: 146, V6: 89
Loopback-flagged events	705	235 repeat Triage + 235 Doctor Reassessment + 235 Documentation Review
Recorded reason	Missing intake paperwork (235 of 235)	Single value; no other reason occurs
Rework delay (rework cases)	mean 35.11, median 35	P10 23, P90 47, max 55+
Mean duration with / without rework	148.63 / 94.86 min	Gap 53.8 min (not like-for-like)
Like-for-like: V4 minus V2	+30.5 min (+29.8%)	Both radiology only
Like-for-like: V6 minus V5	+28.0 min (+19.1%)	Both radiology plus laboratory
Door-to-doctor with / without rework	43 / 43.5 min	No difference: rework happens after the first assessment

Table 10.6 Rework facts

The like-for-like comparison is the fair measure of the cost of rework: it compares a rework pathway with the same pathway without the loop. Both comparisons give about 28 to 30 additional minutes, which matches the "Rework and review" stage in Table 10.3 (about 35 minutes in rework variants against about 6.5 in the corresponding non-rework variants).

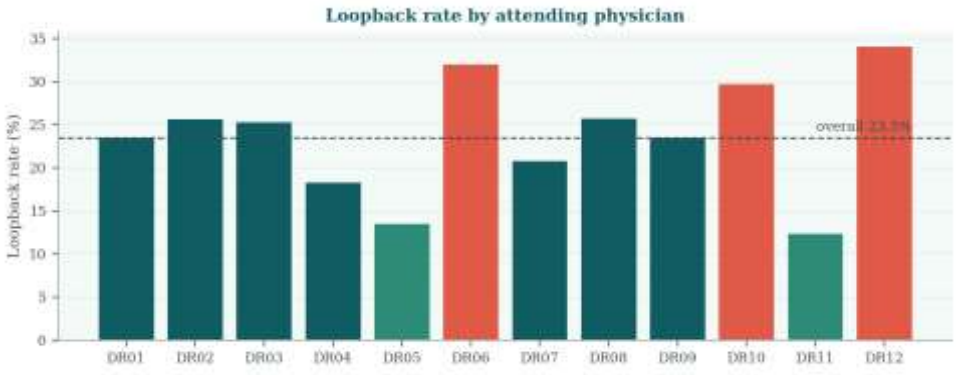


Figure 10.4 Loopback rate by attending physician

Group	Cases	Loopback rate	Mean stay	Door to doctor	Mean acuity
ESI 1	26	23.1%	91.4	14.1	-
ESI 2	144	26.4%	75.8	16.0	-
ESI 3	429	21.7%	112.1	48.9	-
ESI 4	309	24.3%	114.6	49.0	-
ESI 5	92	25.0%	116.3	50.0	-
Age 18-39	306	26.5%	110.1	42.3	3.23
Age 40-59	248	20.6%	106.0	44.4	3.35
Age 60+	446	23.1%	106.5	43.5	3.31

Table 10.7 Rework rate and timing by acuity and age group

Rework is spread evenly across acuity (21.7% to 26.4%) and age (20.6% to 26.5%), so it is a process phenomenon, not a patient-type phenomenon. The physician differences in Figure 10.4 (12.3% to 34.1%) are larger than the other splits but rest on only 64 to 106 cases per physician; a difference of that size can partly arise by chance.

10.5 Triage acuity and the doctor queue

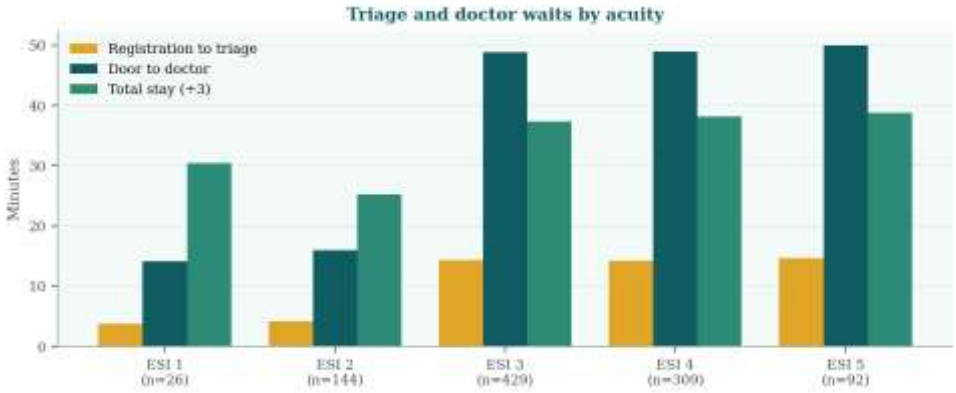


Figure 10.5 Registration-to-triage, door-to-doctor and total stay by acuity

ESI	Cases	Reg to triage	Door to doctor	Assess to discharge	Mean stay	Loopback
1	26	3.8	14.1	77.3	91.4	23.1%
2	144	4.2	16.0	59.8	75.8	26.4%
3	429	14.3	48.9	63.2	112.1	21.7%
4	309	14.3	49.0	65.7	114.6	24.3%
5	92	14.7	50.0	66.3	116.3	25.0%

Table 10.8 Patient flow by acuity

Two regimes are visible. ESI 1 and 2 patients are seen in about 14 to 16 minutes from registration. ESI 3, 4 and 5 patients wait about 49 to 50 minutes with almost no difference between them. In an operating department one would expect a gradient across all five levels; here the queue is effectively two-tiered, so patients at levels 3 to 5 are treated alike in the data.

Mean total stay is slightly longer for less urgent patients (ESI 1: 91.4 min; ESI 3: 112.1 min), giving a positive correlation of 0.232 between the acuity number and duration. This is mostly a consequence of the shorter door-to-doctor time for urgent patients, not of faster downstream care (Table 9.1).

Acuity by pathway

ESI	V1	V2	V3	V4	V5	V6	Total
1	5	4	5	3	6	3	26
2	45	23	24	28	14	10	144
3	116	104	63	55	53	38	429
4	76	68	47	48	43	27	309
5	25	17	10	12	17	11	92

Table 10.9 Number of cases per acuity level and variant

Critical (ESI 1 and 2) patients take every pathway, including rework ones, in roughly the same proportions as other patients. Pathway is therefore not determined by acuity in this data.

10.6 Patient demographics

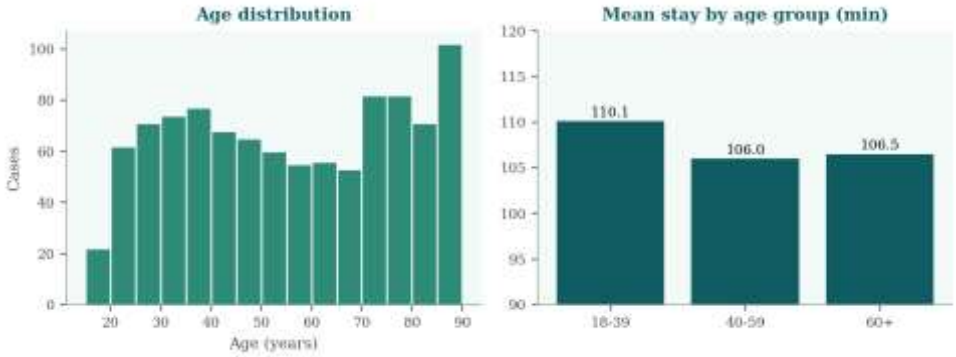


Figure 10.6 Age distribution and mean stay by age group

Age group	Cases	Mean stay	Loopback rate	Mean acuity	Door to doctor
18-39	306	110.1	26.5%	3.23	42.3
40-59	248	106.0	20.6%	3.35	44.4
60+	446	106.5	23.1%	3.31	43.5

Table 10.10 Outcomes by age group

Age is close to uniform between 18 and 90 and has no meaningful relationship with duration (correlation -0.042). The 18-39 group has a slightly longer mean stay (110.1 min) and higher rework rate (26.5%) than the others, but the differences are small relative to the spread of the data (standard deviation 43.5 minutes).

10.7 Temporal patterns

Hour of day



Figure 10.7 Case arrivals by hour

Arrivals per hour range from 28 (12:00) to 60 (16:00). The three busiest hours are 16:00 (60), 15:00 (50), 5:00 (48). The profile is flatter than a real ED (which usually peaks in late morning and evening), another sign the data is synthetic. Event-level counts in N-13 show the same afternoon peak, between 16:00 and 18:00.

Arrival band	Cases	Mean stay	Loopback	Door to doctor	End after midnight
00-05	248	106.1	25.8%	42.7	0
06-11	241	110.9	24.1%	44.3	0
12-17	260	106.2	23.1%	43.6	0
18-23	251	106.9	21.1%	42.9	81

Table 10.11 Six-hour arrival bands

All 81 cases that end after midnight arrived between 18:00 and 23:59, which is a direct consequence of how the flag is defined. Mean stay, rework and door-to-doctor are essentially the same in all four bands.

Day of week, week and month

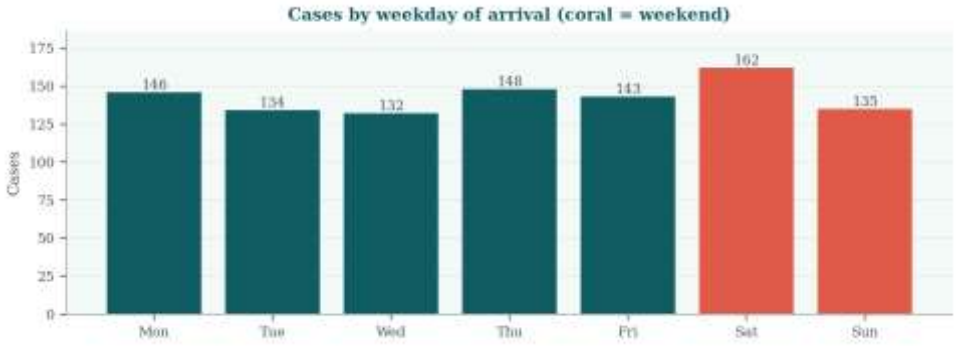


Figure 10.8 Cases by weekday

Weekday	Cases	Mean stay	Loopback	Door to doctor	
Monday	146	107.2	23.3%	43.5	
Tuesday	134	109.8	19.4%	43.7	Quietest day
Wednesday	132	106.6	22.7%	43.4	
Thursday	148	106.5	25%	41.3	
Friday	143	106.7	23.1%	43.6	
Saturday	162	111.0	26.5%	43.7	Busiest day
Sunday	135	104.3	23.7%	44.4	

Table 10.12 Weekday profile

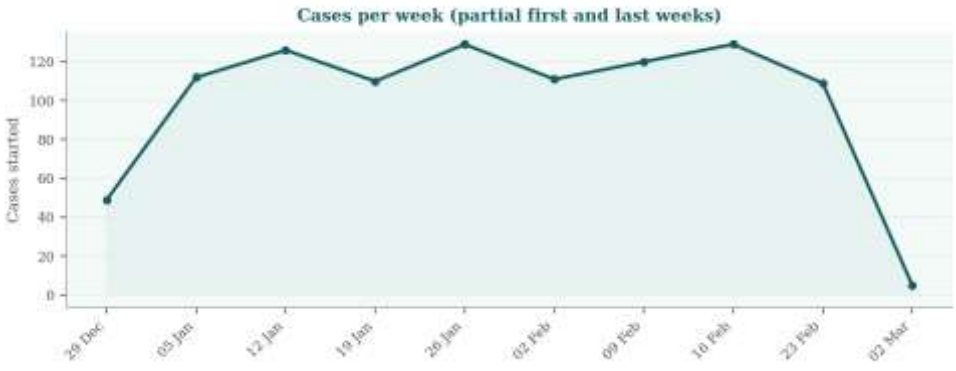


Figure 10.9 Weekly case volume

Weekends carry 297 cases (29.7%) and have a mean stay of 107.9 minutes against 107.3 on weekdays, a gap of 0.6 minutes that is too small to matter. Monthly volumes are January 508, February 467 and March 25 (partial month: 1 to 2 March). Daily volume averages 16.39 cases with a range of 5 to 25.

Shift

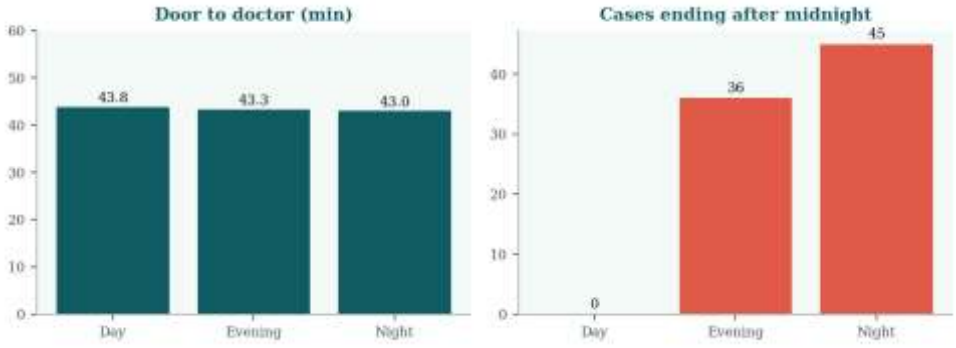


Figure 10.10 Door-to-doctor time and midnight-crossing cases by shift

Shift	Cases	Mean stay	Door to doctor	Loopback	End after midnight
Day 07-15	312	110.6	43.8	25.6%	0
Evening 15-23	358	105.1	43.3	21.5%	36
Night 23-07	330	107.1	43.0	23.6%	45

Table 10.13 Shift comparison

Shift does not change door-to-doctor time (43.0 to 43.8 minutes) or the rework rate in any meaningful way. Its operational relevance is boarding: only Evening (36) and Night (45) arrivals cross midnight. Night-shift cases that do so are those arriving between 23:00 and 23:59.

10.8 Physician analysis

Doctor	Cases	Share	Mean stay	Loopback	Day	Evening	Night
DR01	102	10.2%	101.0	23.5%	110.2	99.8	94.7
DR02	86	8.6%	109.6	25.6%	111.6	99.9	116.2
DR03	79	7.9%	105.2	25.3%	111.7	112.3	90.4
DR04	82	8.2%	114.1	18.3%	114.5	115.1	112.7
DR05	89	8.9%	102.5	13.5%	101.7	103.8	102.2
DR06	75	7.5%	109.2	32.0%	112.4	106.2	109.1
DR07	106	10.6%	104.7	20.8%	109.2	106.2	97.9
DR08	74	7.4%	108.9	25.7%	104.5	105.3	119.6
DR09	85	8.5%	108.1	23.5%	117.9	92.2	113.2
DR10	64	6.4%	108.9	29.7%	117.9	97.2	111.7
DR11	73	7.3%	103.6	12.3%	104.2	100.5	105.6
DR12	85	8.5%	116.4	34.1%	116.2	115.6	117.8

Table 10.14 Physician volume, mean stay, loopback rate and mean stay by shift (minutes)

Case volume per physician ranges from 64 (DR10) to 106 (DR07). Mean acuity of their patients is nearly constant (3.23 to 3.37), so physicians face similar case mixes. Mean stay ranges from 101.0 to 116.4 minutes. The Power BI report "Doctors and Workloads" presents the same breakdown as a bar chart and a physician-by-shift matrix.

> **WARNING | Interpret physician differences with care**
> Splitting 1,000 cases over 12 physicians and 3 shifts leaves cells of about 20 to 40 cases. Differences of 10 to 20 minutes in a cell can arise from chance alone given the standard deviation of 43 minutes. Use these tables to decide where to look, not to compare individuals.

10.9 Correlation analysis

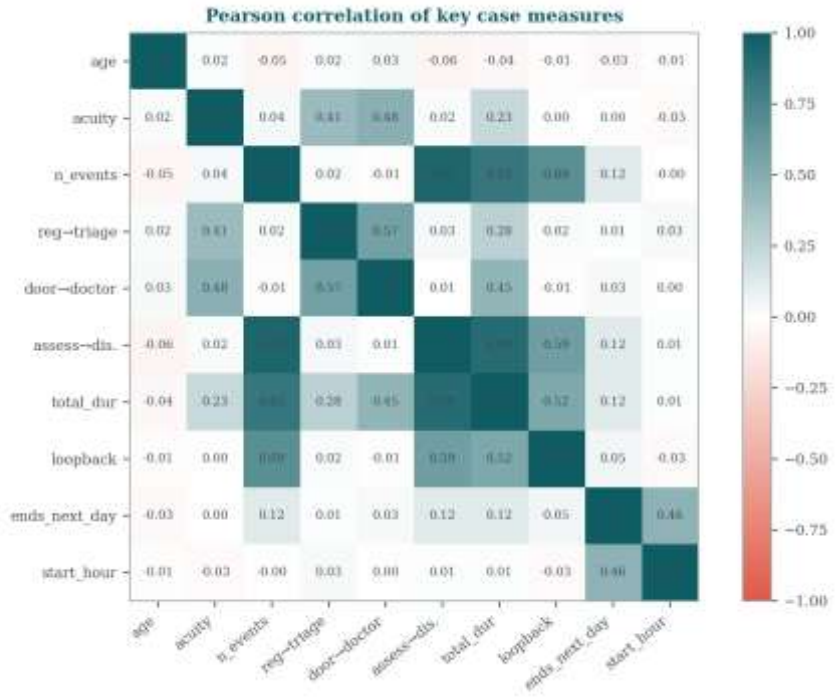


Figure 10.11 Pearson correlations between key case measures

Three relationships stand out. Total duration correlates most strongly with assessment-to-discharge time (it contains it), with the number of events (more steps, longer stay), and with the loopback flag (r = 0.52). Acuity has a weak positive correlation with duration (0.23). Age has no correlation with anything. The ends-next-day flag correlates with arrival hour (0.46) because only late arrivals can cross midnight, and only weakly (0.12) with duration.

Acuity correlates with registration-to-triage time (0.41) and door-to-doctor time (0.48), which is the two-tier queue described in Section 10.5. Door-to-doctor correlates moderately with total duration (0.45), while assessment-to-discharge time (0.90) dominates it, and the two components are almost uncorrelated with each other (0.01).

10.10 Transition-level view

Table 8.2 lists all 16 transitions with percentiles. Three observations matter for operational follow-up. First, the three slowest transitions (Lab Collection to Lab Result, Triage to Doctor Assessment, X-Ray to Radiology Result) have 90th percentiles of 48, 55 and 33 minutes against means of 32.6, 30.7 and 22.4, so one in ten patients waits much longer than the average suggests. Second, seven of the 16 transitions have a mean below 13 minutes. Third, the maximum for Triage to Doctor Assessment is 60 minutes, so no patient waited longer than an hour for a doctor in this dataset.

10.11 Summary of key measurements

Measure	Value	Where documented
Cases / events / transitions	1,000 / 6,964 / 5,964	Section 6.1
Mean / median / P90 duration	107.5 / 105 / 165 min	Table 10.4

Measure	Value	Where documented
Mean door to doctor	43.4 min (P90 69)	Table 10.4
Critical vs standard door to doctor	14-16 vs 49-50 min	Table 10.8
Slowest transition	Lab Collection to Lab Result, 32.6 min	Table 8.2
Cases with rework	235 (23.5%)	Table 10.6
Rework cost, like for like	+28 to +30 min	Table 10.6
Cases ending after midnight	81 (8.1%), all arriving 18:00-23:59	Table 10.11
Loopback rate range across physicians	12.3% to 34.1%	Figure 10.4
Correlation of the loopback flag with duration	r = 0.52	Figure 10.11

11 Dashboards and Visualisations

Three presentation channels exist: static charts for documents, a Streamlit application for exploration, and four Power BI reports for managers. This chapter specifies each and explains how to read it.

11.1 Static charts (M-14)

Chart file	Content	Source data
`activity_frequency_chart.png`	Horizontal bars: events per activity	`activity_frequency_summary.csv`
`patient_pathway_chart.png`	Bars: patients per pathway, with pathway labels shortened (P1 to P6)	`patient_journey_sequence_summary.csv`
`workflow_elapsed_time_chart.png`	Bars: ten transitions with highest mean elapsed minutes	`workflow_performance_summary.csv`
`workflow_visualization_pathway_details.csv`	Key from short pathway labels to full activity sequences	Generated alongside the charts

The script locates columns with a `find_column` helper that tries several candidate names, so a renamed column in an input CSV does not break chart generation. It exits with a clear message when an input file is missing.

11.2 Streamlit dashboard (M-15)

Layout

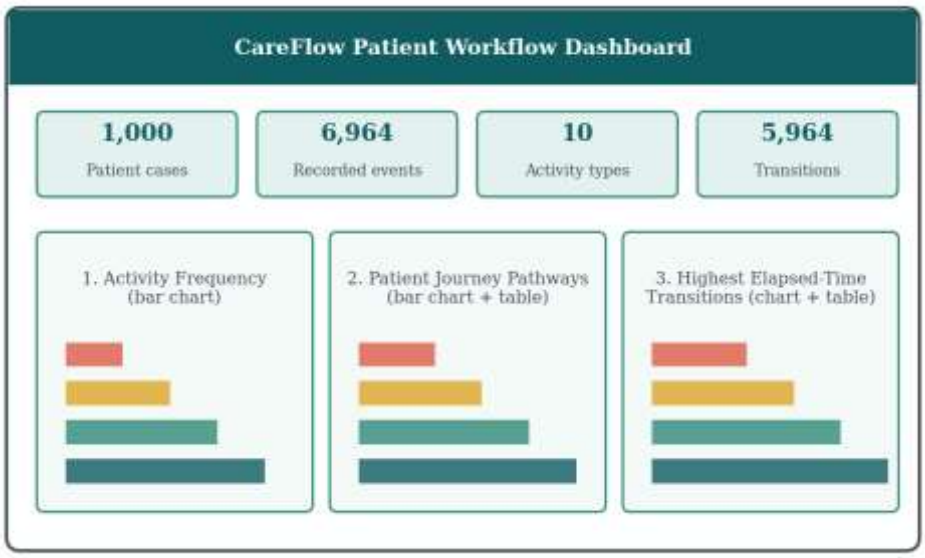


Figure 11.1 Layout of the Streamlit dashboard (schematic)

Specification

Framework	Streamlit; page configured with wide layout and a hospital icon in the title
Data loading	A `load_data()` function reads the three summary CSVs and returns data frames; missing files display an error and stop the page
Column tolerance	`find_column(df, candidates)` matches columns case-insensitively across alternative names
KPI cards	Patient cases, recorded events, activity types, transitions, shown with `st.metric` in four columns

Section 1	Activity frequency: bar chart of event counts per activity
Section 2	Patient journey pathways: bar chart of patients per pathway and a table with full sequences
Section 3	Highest average elapsed-time transitions: bar chart and table (count, mean, median)
State	Stateless; no filters or user input; every refresh re-reads the CSVs
Command	`streamlit run src/careflow_interactive_dashboard.py`
Default address	http://localhost:8501

How to read it

- The KPI cards confirm the data volume at a glance: they should read 1,000, 6,964, 10 and 5,964.
- In section 2 the longest bar is the fast-track pathway; the other five pathways include diagnostics or rework.
- In section 3, transitions with higher mean elapsed time are review candidates. Read them together with the count column: a slow transition that few patients experience matters less than one every patient passes through (Triage to Doctor Assessment has 1,000 cases).

11.3 Power BI reports

Four single-page reports share one visual language: a dark title bar, a banner reading "Synthetic data, 1 Jan to 2 Mar 2026", and slicers for acuity, shift, weekend and age group. They connect to `careflow_cases_imputed.csv` and `careflow_events_scaled.csv`.

Report	Page	Visuals
`CareFlow_Emergency_Dashboard.pbix`	Overview	Cards: Total Cases, Avg Duration, Avg Door to Doctor, Loopback %. Charts: cases per week, average duration by age group, average duration by shift, top 5 pathway variants.
`CareFlow_Process_and_Bottlenecks.pbix`	Process and bottlenecks	Average wait before each activity (bar); top 5 variants (bar).
`CareFlow_Doctors_and_Workloads.pbix`	Doctors and workload	Average duration by doctor (bar); doctor by shift average duration (matrix).
`CareFlow_Loopbacks_and_Outliers.pbix`	Loopbacks and outliers	Cards: Loopback Cases, Loopback %, Outlier Cases, Top Loopback Reason. Charts: loopback % by acuity; age versus duration scatter coloured by acuity; page navigator.

Slicers and interaction

All four slicers cross-filter every visual on the page. Selecting ESI 1 and 2 shows the critical cohort; selecting the Night shift isolates night-time patterns; the age-group slicer reproduces Table 10.10. Because each report is one page, the navigator on the loopback page links to the other three.

Measures referenced

Measure name	Interpretation
Total Cases	Count of rows in the case table after slicer filters
Avg Duration	Mean of `total_duration_min`
Avg Door to Doctor	Mean of `door_to_doctor_min`
Loopback %	Share of cases with `loopback_flag` = 1
Outlier Cases	Count of cases with `duration_outlier_flag` = true (1 in the data)
Top Loopback Reason	Most frequent value of `loopback_reason` (Missing intake paperwork)
Avg wait before each activity	Inferred from the visual title and fields: mean of `minutes_since_prev` by `activity_name` in the event table

> **NOTE | How the reports were reviewed**
> The `.pbix` files were inspected structurally (pages, visual types, titles, fields). Measure formulas and rendered appearance were not examined because the files were not opened in Power BI Desktop. The measure interpretations above follow the names and fields found in the report definitions.

11.4 Stakeholder summary

`STAKEHOLDER_SUMMARY.md` is a one-page executive note organised under three headings: bottleneck isolation, workload distribution and data integrity. It proposes automated routing for high-acuity patients, shift schedules aligned with intake peaks and real-time process-mining dashboards. Several statements in it need to be restated before circulation (issues I-02, I-06 and I-07).

11.5 Visual design guidance

- Show counts next to rates so small groups are not over-read.
- Label time measures as "elapsed" unless a column explicitly measures waiting.
- Keep the "Synthetic data" banner on every page so findings are not mistaken for real hospital performance.
- Use a sequential colour for magnitude and a contrasting colour only for the rework path, as in Figure 8.2.

12 Verification and Validation

12.1 Approach

Verification asks whether each component does what it claims; validation asks whether the numbers it produces are right. The project has no automated test suite, so assurance rests on four layers.

#	Layer	Description
1	Defensive coding	Every script validates columns, nulls and timestamps and raises an error otherwise.
2	Built-in checks	The preparation module writes 16 PASS/FAIL checks (Section 7.8).
3	Cross-module reconciliation	Totals (1,000 cases, 6,964 events, 5,964 transitions, 235 rework cases) agree across independent modules and notebooks.
4	Independent recomputation	For this document, headline figures were recomputed with pandas directly from the shared CSV files and compared with the published ones.

NOTE | What was and was not executed
The scripts and notebooks were read and their outputs inspected, and the published numbers were recomputed independently from the data files. The analysis scripts themselves were not re-executed here, because the full event log is only available in the `akshar` branch and the scripts expect it in a different folder (issue I-10).

12.2 Reconciliation of published figures

Figure	Published	Recomputed	Status
Cases / events / activities	1,000 / 6,964 / 10	1,000 / 6,964 / 10	Agree
Transitions / transition types	5,964 / 16	5,964 / 16	Agree
Pathway variants and counts	267, 216, 149, 146, 133, 89	Same	Agree
Cases with loopback	235 (23.5%)	235 (23.5%)	Agree
Loopback-flagged events	705	705	Agree
Mean total duration	107.5 min	107.50 min	Agree
Mean door to doctor	43.4 min	43.36 min	Agree
Slowest transition	Lab Collection to Lab Result 32.56	32.56	Agree
Acuity correlation with duration	0.232	0.232	Agree
Critical cases (ESI 1-2)	170	170	Agree
Cases ending after midnight	81	81	Agree
Train / test cases and cutoff	800 / 200; 2026-02-18 11:00	Same	Agree
Duration outliers	80 (8.0%)	1 flagged; 1 by IQR on total duration	Disagree (I-01)
Evening door-to-doctor higher	"disproportionately higher"	43.8 / 43.3 / 43.0 (Day / Evening / Night)	Disagree (I-04)
Day loopback rate	26.0%	25.6%	Minor difference (I-05)
Midnight crossings Evening / Night	37 / 44	36 / 45	Minor difference (I-05)
Average wait time	64.1 and 107.5 both quoted	64.14 = assessment to discharge; 107.50 = total duration	Ambiguous (I-03)

12.3 Test cases

The table lists the behaviours that matter most, how they were confirmed and how they should be automated. IDs TC-01 to TC-12 are suggested names for a future pytest suite.

ID	Behaviour	Evidence today	Suggested automated test
TC-01	Script raises when a required column is absent	Code review (`ValueError`)	Drop one column from a small fixture and expect the error
TC-02	Script raises on null values or bad timestamps	Code review	Insert NaN / "not a date" and expect the error
TC-03	Number of transitions = events - cases	5,964 = 6,964 - 1,000	Assert on any fixture
TC-04	No negative elapsed time survives	All 5,964 elapsed times are non-negative	Fixture with reversed timestamps is filtered
TC-05	Chronological split has no overlap	Check 1 of the log	Assert max(train) <= min(test)
TC-06	Structural fields are zero exactly when flag is 0	Checks 2 to 4	Assert equivalence on fixture
TC-07	Scalers fitted on training rows only	Check 8 to 10	Change test rows; parameters must not change
TC-08	Inverse transform recovers original	Check 12	Round-trip within 1e-9
TC-09	Model-ready features contain no target columns	Code review of `feat` list	Assert no column starts with `target_` in feature list
TC-10	Variant counts sum to total cases	1,000	Assert sum equals unique cases
TC-11	Dashboard starts with the three CSVs present	Manual	Streamlit AppTest smoke test
TC-12	Dashboard reports missing file clearly	Code review	Remove a CSV and assert message

12.4 Statistical sanity checks

- **Scaling:** training means of standardised features are zero to within 1.3e-16, training standard deviations are one, and Min-Max training values span exactly [0, 1].
- **Interval consistency:** door-to-doctor (43.36) equals registration-to-triage (12.62) plus triage-to-doctor (30.74) to rounding.
- **Component consistency:** mean total (107.50) equals door-to-doctor (43.36) plus assessment-to-discharge (64.14).
- **Rework consistency:** the stage table (Table 10.3) reproduces the like-for-like rework cost (about 28 to 30 minutes) found by comparing variant means.
- **Distribution:** 0 of 1,000 observed values missing; durations bounded between 12 and 234 minutes; no negative values.

12.5 Residual risk

Because the data is synthetic and strongly structured (for example only two door-to-doctor regimes, one loopback reason, uniform age), results validate the pipeline rather than hospital behaviour. Before the methods are used on real data, expect noisier distributions, many more variants, missing and out-of-order timestamps, and multiple departments.

13 Installation and Operation Guide

13.1 Prerequisites

Item	Requirement
Operating system	Windows, macOS or Linux (notebooks were developed on Windows)
Python	3.10 or later
Packages	pandas, numpy, scikit-learn, pm4py, matplotlib, seaborn, plotly, streamlit, jupyter, tabulate (needed by `to_markdown` in M-16)
Graphviz	Required by PM4Py to render the DFG (M-04)
Power BI Desktop	Windows only; needed to open the `.pbix` files
Data	The cleaned case and event tables, or the eight artefacts from the `akshar` branch

13.2 Installation

Setup commands

```
git clone https://github.com/akshar059/Infotact_P1_CareFlow.git
cd Infotact_P1_CareFlow
git checkout prachi           # or akshar / sagar
python -m venv venv
venv\Scripts\activate         # macOS/Linux: source venv/bin/activate
pip install pandas numpy scikit-learn pm4py matplotlib seaborn plotly streamlit jupyter tabulate
```

13.3 Execution order

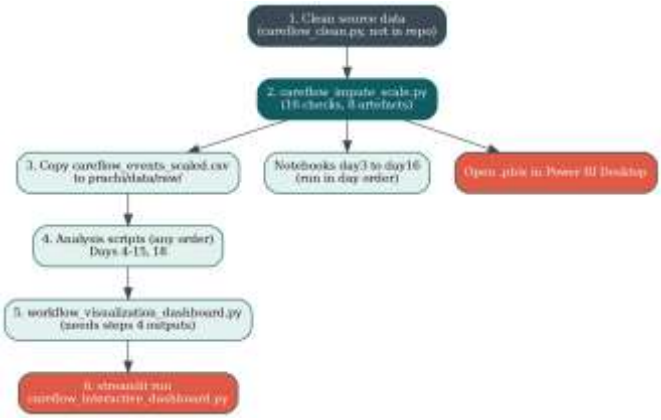


Figure 13.1 Recommended execution order

13.4 Running each component

Component	Procedure
Preparation (`akshar`)	`python careflow_impute_scale.py <source_dir> <output_dir>`. The source directory must contain `careflow_cases_clean.csv` and `careflow_events_clean.csv`. Expect the line "PASS: 16 FAIL: 0" and a shapes line (1000x32, 1000x33, 1000x33, 6964x27, 1000x41).
Copy the event log	Copy `careflow_events_scaled.csv` from the preparation output into `prachi/data/raw/` before running any M-module other than M-01, M-17 and M-18.
Analysis modules (`prachi`)	From the repository root: `python src/<script>.py`. Each prints a summary and writes to `data/processed/`. M-14 and M-15 need the outputs of M-07, M-09 and M-12 first.
Static charts	`python src/workflow_visualization_dashboard.py` writes three PNG files to `docs/`.
Streamlit	`streamlit run src/careflow_interactive_dashboard.py`; open the printed local URL.
Notebooks (`sagar`)	`jupyter notebook`, open `notebooks/`, and run "Restart and Run All" for each notebook in day order N-03 to N-16.
Power BI	Open a `.pbix` file. If the data source path is wrong choose Transform data, Data source settings, Change source, and point to the two CSVs.

13.5 Expected console output

process_bottleneck_analysis.py (M-11)

```
=== PROCESS BOTTLENECK CANDIDATE ANALYSIS ===
Total patient cases: 1000
Total recorded events: 6964
Valid consecutive transitions: 5964
Unique transition types: 16
```

workflow_improvement_recommendations.py (M-13)

```
=== WORKFLOW IMPROVEMENT RECOMMENDATIONS ===
Patient cases: 1000     Recorded events: 6964
Valid transitions analyzed: 5964     Recommendations generated: 16
```

13.6 Troubleshooting

Symptom	Likely cause	Action
`FileNotFoundError` for `careflow_events_scaled.csv`	Event log not copied into `data/raw/`	Copy the file from the preparation output
`ValueError: Missing columns`	Wrong file (for example the 16-event sample log) or renamed columns	Check the file header against Section 6.4
`FileNotFoundError: Required analysis file not found`	M-14 or M-15 run before M-07, M-09, M-12	Run those three modules first
PM4Py fails to render the DFG	Graphviz not installed or not on PATH	Install Graphviz and restart the shell
`ImportError: tabulate`	M-16 uses `DataFrame.to_markdown`	`pip install tabulate`
Streamlit page blank or shows error	Processed CSVs missing	Check `data/processed/` contains the three summary files
PBIX shows "Couldn't load the data"	Source path differs on this machine	Update the data source path
Preparation script shows FAIL lines	Input changed or corrupted	Read the detail column of `scaling_checks.csv`
Numbers differ from this document	Different input data	Compare event and case counts first

13.7 Frequently asked questions

Why are some interval columns zero? The step did not occur for that patient. Always filter on the indicator flag before averaging (Section 7.4).

Why does Triage have 1,235 events but 1,000 patients? 235 patients are triaged twice as part of the rework loop.

Can the dashboard be filtered? The Streamlit app cannot; the Power BI reports have four slicers.

Can I use the feature store to predict stay length on a new patient? Yes, after transforming the new row with the parameters in `scaler_parameters.json`. Only the arrival-time columns of Section 7.7 may be used as inputs.

Is any real patient data included? No. The data is synthetic.

14 Maintenance and Extension Guide

14.1 Adding a new data period or dataset

1. Produce cleaned case and event tables with the same columns as Section 6.3 and 6.4.
2. Run the preparation module on the new files. Read the 16-check log; investigate any FAIL.
3. Copy the new `careflow_events_scaled.csv` to `prachi/data/raw/` and re-run M-02 to M-16, then the chart script.
4. Re-run notebooks N-03 to N-16 in order and refresh the Power BI reports.
5. Compare headline numbers with Section 10.11 and update the hand-written write-ups in `docs/`.

14.2 Adding a new activity

- Add the activity to the upstream cleaning step and to the event table (name, category).
- The generic modules (M-02 to M-13, M-16) discover activities from the data and need no change.
- `recommendation_map` in M-13 is hand-written; add an entry if a new transition deserves specific advice.
- Update the activity catalogue (Section 2.1) and the transition matrix figure.

14.3 Extending the feature store

Any new feature must be knowable at arrival. Before adding a column, ask: "Would a triage nurse have this value at the moment the patient is registered?" If the answer is no, it belongs in an analysis table, not the feature store. When a feature is added, add it to `feat` (so the NaN check covers it), fit its scaler on training rows only, and record its parameters in `scaler_parameters.json`.

14.4 Code quality improvements

#	Improvement	Benefit
1	Move load/validate/sort into a shared `careflow_common.py`	Removes duplicated code in 16 scripts; one place to fix defects
2	Wrap each script body in `main()` with `if __name__ == "__main__"`	Allows import and unit testing; M-16 already does this
3	Replace hard-coded activity/transition maps with configuration files	Easier edits without touching code
4	Add pytest tests TC-01 to TC-12	Regression protection when data or code changes
5	Generate `docs/*.md` from the CSVs	Prevents drift between write-ups and outputs
6	Make the preparation module exit non-zero on any FAIL	Stops downstream steps on a bad run
7	Add a `requirements.txt` with pinned versions	Reproducible environments
8	Add a top-level `run_all.py`	One command reproduces everything

14.5 Version control practice

- Keep `main` as the consolidated, reviewed state and merge member branches through pull requests with at least one reviewer.
- Commit generated artefacts only when they are small summaries; keep the full event log in one agreed location and document its path.
- Tag a release when a dataset version and its documents agree, for example `v1.0-data-2026-03`.

- Use consistent folder layouts across branches so merging does not require moving files.

14.6 Privacy and security considerations

The present data is synthetic, but the design anticipates real data. Case identifiers should be pseudonymous, never medical record numbers. Dates and ages are quasi-identifiers; before publishing results derived from real data, check that small groups (for example one physician on one shift) cannot identify individuals. The dashboards read summary CSVs only, which limits exposure; keep it that way. Do not commit real event logs to a public repository.

14.7 Risk register

#	Risk	Likelihood	Impact	Mitigation
1	Findings from synthetic data are mistaken for real performance	Medium	High	Keep the synthetic-data banner and wording in every output
2	Elapsed time is read as waiting time	High	Medium	Use "elapsed" consistently; confirm timestamp semantics
3	Write-ups drift from CSV outputs	Medium	Medium	Generate documents from code or add checks
4	Scripts cannot be re-run because inputs are not committed	High	Medium	Document data location; add the cleaning step
5	Physician differences are over-interpreted	Medium	High	Add confidence intervals; case-mix adjustment
6	Model leakage when a future model is built	Medium	High	Restrict to the feature store; use time-ordered validation
7	Real data has many more variants than six	High	Medium	Filter to top-N variants and group the long tail

15 Known Issues and Change Requests

Issues were found while consolidating the four branches and checking published figures against the data (Chapter 12). Severity follows a simple scale: **High** means a published conclusion is wrong or misleading; **Medium** means a figure or label needs correction; **Low** means a cosmetic or process improvement.

15.1 Issue log

ID	Severity	Where	Observation	Evidence	Proposed fix
I-01	High	`sagar` README, N-16, final metrics CSV	Reports 80 duration outliers (8.0%).	N-16 falls back to `int(total_cases * 0.08)` when its expected column is missing. N-11 found 0 outliers on assessment-to-discharge; the dataset flags 1 case and an IQR rule on total duration also gives 1.	Replace by the computed value; compute rather than hard-code.
I-02	High	`sagar` README, N-16, final metrics CSV	Bottleneck pairs named "Triage to Lab Result" (32.6) and "Registration to Doctor Assessment" (30.7).	Measured transitions are Lab Collection to Lab Result (32.56) and Triage to Doctor Assessment (30.74). Four values are typed into the notebook. "Doctor Assessment to Discharge Decision, 15.1" matches no measured transition (nearest 14.8 and 15.5).	Rename; derive the table from the transition summary.
I-03	Medium	N-10, N-09, READMEs	"Average wait time" is 64.1 in one place and about 107.5 in another.	64.14 is the mean assessment-to-discharge interval; 107.50 is the mean total duration. In N-09 the column `total_wait_time_min` is a copy of `total_duration_min` (r = 1.0).	Use distinct names: "assessment-to-discharge" and "total duration".
I-04	High	`main` README	States that evening admissions have disproportionately higher door-to-doctor times.	Door-to-doctor is 43.8 (Day), 43.3 (Evening), 43.0 (Night) minutes. The shift effect concerns midnight crossings, not waits.	Remove or restate.
I-05	Low	`main` README shift table	Day loopback 26.0%; midnight crossings 37 (Evening) and 44 (Night).	Recomputed: 25.6%; 36 and 45 (total 81 unchanged).	Update the table.
I-06	Medium	`STAKEHOLDER_SUMMARY.md`	Mentions Emergency-to-ICU handoffs and nursing allocation.	The data has one department and no ICU events or staff-role fields.	Rewrite to match the data.
I-07	Medium	N-13, N-14, N-15, final CSVs	Summary artefacts are placeholders.	`day13_peak_loads_summary.csv` marks every hour "Normal"; the importance table has one feature (acuity = 1.0); executive summary text is generic.	Regenerate from data with a defined peak threshold and several features.
I-08	Low	`sagar` README	Lists notebook names that do not exist.	Files are `day3_` to `day16_`; README lists `day01_`, `day11_outlier_detection` and similar.	Align names.
I-09	Low	M-13 output	Recommendation 1 is "Further review" ahead of the six "High investigation priority" rows.	The sort key is the priority text, sorted alphabetically.	Sort by an explicit priority rank.
I-10	Medium	`prachi` branch	Scripts read `data/raw/careflow_events_scaled.csv`, which is not committed.	Only the 16-event sample is in the branch, so M-02 to M-16 cannot run without copying the file in.	Document the data location or add the file.
I-11	Medium	`akshar` branch, `main` README	Preparation needs `careflow_cases_clean.csv` and `careflow_events_clean.csv` from `careflow_clean.py`, which is absent. README run command and file tree give different script paths; README says "15-point" log, shipped log has 16 checks.	Cannot run end to end from the repository.	Add the cleaning script; align paths and counts.

ID	Severity	Where	Observation	Evidence	Proposed fix
I-12	Low	Preparation module	Imputation benchmark was run but no genuine missing values exist.	Only structural zero-fill was applied.	State this explicitly in reports (done in Section 7.4).
I-13	Low	Several write-ups, file names	"Waiting time" used for elapsed time.	Elapsed time includes activity duration and timestamp meaning is undocumented.	Use "elapsed time"; confirm timestamp semantics.
I-14	Low	N-13	Column `patient_count` counts events, not patients.	Hourly values are about 250 to 350, whereas case arrivals per hour are 32 to 60.	Rename the column or count unique cases.

Table 15.1 Issue log

15.2 Corrected statements for reuse

The following wording replaces the affected statements wherever they appear in READMEs, the stakeholder summary or presentations.

Issue	Recommended wording
I-01	One of 1,000 cases is flagged as a duration outlier (IQR rule on total duration). No outliers are detected on assessment-to-discharge time.
I-02	The slowest transitions by mean elapsed time are Lab Collection to Lab Result (32.6 min), Triage to Doctor Assessment (30.7 min) and X-Ray to Radiology Result (22.4 min).
I-03	Mean total duration is 107.5 minutes, of which 43.4 minutes is door-to-doctor time and 64.1 minutes assessment-to-discharge time.
I-04	Door-to-doctor time is similar across shifts (43.0 to 43.8 minutes). Evening and night arrivals are more likely to remain after midnight (81 cases).
I-06	The data covers a single Emergency Department. It contains no ICU or nursing-allocation information.

16 Predictive Modelling Plan

The feature store was built so that modelling can start immediately. This chapter documents the intended approach so that the work is consistent when it begins. No models have been trained yet.

16.1 Prediction tasks

ID	Task	Target column	Use if successful
P1	Predict length of stay at arrival (regression)	`target_total_duration_min`	Bed planning, expected-wait messages to patients
P2	Predict whether rework will occur (binary)	`target_loopback`	Prompt for intake-paperwork completeness at registration
P3	Predict whether the case will cross midnight (binary)	`target_ends_next_day`	Early boarding and staffing alerts for the evening shift

16.2 Validation design

- Use the supplied chronological split (800 train, 200 test) for the final report.
- For tuning, use expanding-window cross-validation on the training period (for example five folds, each trained on earlier weeks and tested on the next), never random folds.
- Report results on the 200 test cases once, after all choices are made.
- Compare every model with a trivial baseline: the training mean (regression) and the training prevalence (classification).

16.3 Candidate models and metrics

Task	Candidate models	Metrics
P1 regression	Baseline mean, Ridge, Random Forest, Gradient Boosting, Huber regression	MAE, median absolute error, RMSE, R-squared; also, MAE within acuity groups
P2 classification	Prevalence baseline, Logistic Regression, Random Forest, Gradient Boosting	PR-AUC, ROC-AUC, balanced accuracy, recall at fixed precision, Brier score
P3 classification	Same as P2; hour-of-day sine/cosine features are central	PR-AUC, recall, calibration curve

16.4 Expectations from the data

The correlation analysis (Section 10.9) sets realistic expectations before any model is run.

- **P1.** The only arrival-time feature correlated with duration is acuity ($r = 0.23$), which would explain about 5% of the variance by itself. Age, hour, weekday, shift and physician show no relationship. A model should therefore beat the mean only modestly, and a large R-squared would be a red flag for leakage.
- **P2.** Rework has no correlation with acuity (0.00), age (-0.01) or hour (-0.03), and rates by physician differ by chance-level amounts or by genuine physician effects. A classifier may be unable to beat the 23.5% baseline. If so, that is a valid finding: rework is not predictable from patient characteristics and is better addressed as a process problem.
- **P3.** The midnight flag is determined mainly by arrival hour ($r = 0.46$) and by duration; arrival hour plus expected duration should predict it well, and its sine/cosine features are designed for exactly this.

> **WARNING | Leakage traps**
> Do not add `n_events`, `has_xray`, `has_lab`, `variant_id`, any interval column or `loopback_*` to a model that is meant to predict at arrival. They are known only after the patient has been seen. Doing so produces very high scores (n_events alone has r = 0.83 with duration) and no usable model.

16.5 Deliverables for the modelling phase

- Notebook or script that fits the baseline and candidate models with time-ordered validation.
- A model-evaluation table for the final report (the pending "model evaluation results" deliverable).
- Permutation importance of arrival features, replacing the single-feature table of N-14.
- A short model card describing training period, features, metrics, limitations and intended use.

17 Operational Recommendations and Monitoring

The recommendations below follow from the measurements in Chapter 10. They are suggestions for investigation. Elapsed time alone does not prove a bottleneck or its cause, and the data is synthetic; before acting on any item, confirm it on real data and with the staff who run the process.

17.1 Recommendation register

ID	Recommendation	Supporting evidence	Owner (suggested)	Priority
RC-1	Investigate laboratory turnaround: transport, analyser queue, result time-stamping	Lab Collection to Lab Result mean 32.6, P90 48 min (371 cases)	Laboratory manager	High
RC-2	Review the standard-acuity queue by hour; test a fast track for ESI 3 to 5	Door to doctor 49-50 min for ESI 3-5 against 14-16 for ESI 1-2; P90 69	ED operations manager	High
RC-3	Introduce an intake-completeness check at registration or triage	All 235 rework cases cite missing intake paperwork; rework costs 28-30 min	Clinical lead, registration	High
RC-4	Review imaging turnaround and reporting workflow	X-Ray to Radiology Result mean 22.4 min (584 cases)	Radiology manager	Medium
RC-5	Plan beds and staffing for late-evening arrivals	81 cases cross midnight, all arriving 18:00-23:59	Charge nurse	Medium
RC-6	Review physician variation using adjusted comparisons	Loopback 12.3% to 34.1% across 12 physicians	Clinical lead	Medium
RC-7	Confirm what each timestamp represents	Elapsed time mixes work and waiting	Data owner	Medium
RC-8	Review documentation and discharge approval steps	Documentation Review to Discharge mean 15.5 min (216 cases)	Records manager	Low

Table 17.1 Recommendation register

17.2 Monitoring metrics

A small set of measurements, tracked weekly with the same definitions as this document, will show whether changes work. Baselines come from the current data.

Metric	Definition	Baseline	Target direction	Source
Median door to doctor (ESI 3-5)	Registration to Doctor Assessment	49 min	Down	Case table
P90 door to doctor (all)	90th percentile	69 min	Down	Case table
Lab turnaround P50 / P90	Lab Collection to Lab Result	32 / 48 min	Down	Event log
X-ray turnaround P50 / P90	X-Ray to Radiology Result	22 / 33 min	Down	Event log
Rework rate	Share of cases with loopback	23.5%	Down	Case table
Rework cost	Rework minus matched non-rework variant	28-30 min	Down	Variant means
Overnight boarding	Cases ending after midnight	81 (8.1%)	Down	Case table
Total duration P50 / P90	`total_duration_min`	105 / 165 min	Down	Case table

Table 17.2 Key performance indicators and current baselines

17.3 Prioritisation

Action	Expected impact	Effort	Rationale
Intake-completeness check (RC-3)	High	Low	Removes the only recorded cause of rework (23.5% of cases, about 28-30 minutes each)
Standard-acuity fast track (RC-2)	High	Medium	Affects 83% of patients and the largest single wait (about 49 min)
Lab turnaround review (RC-1)	Medium	Medium	Slowest transition, 37% of patients
Late-evening boarding plan (RC-5)	Medium	Low	Predictable from arrival time
Timestamp semantics (RC-7)	Enabler	Low	Makes every other elapsed-time number interpretable

17.4 Next steps for the project

1. Correct the documentation items in Chapter 15.
2. Add the cleaning script and data location so every module can be re-run (I-10, I-11).
3. Execute the modelling plan (Chapter 16) and complete the model-evaluation deliverable.
4. Write the project report and presentation using the verified figures in this document.
5. Validate with real, de-identified data before proposing operational change.

Appendix A File Inventory

A.1 Branch akshar

File	Shape	Description
`careflow_impute_scale.py`	180 lines	Preparation and validation module
`careflow_cases_imputed.csv`	1000 x 32	Complete case table with flags
`careflow_cases_standardized.csv`	1000 x 33	Z-score case features
`careflow_cases_normalized.csv`	1000 x 33	Min-Max case features
`careflow_events_scaled.csv`	6964 x 27	Event log with scaled times
`careflow_model_ready_scaled.csv`	1000 x 41	Arrival-time feature store
`imputation_benchmark.csv`	9 x 5	NRMSE by method and column (8 columns + mean row)
`scaling_checks.csv`	16 rows	Validation log
`scaler_parameters.json`	-	Training-fit constants
`project assignment letter.pdf`	1 file	Assignment ref ITS/DA/1176
`README.md`	1 file	Branch documentation

A.2 Branch prachi

File group	Contents
`src/` (18 scripts)	M-01 to M-18 (Chapter 8)
`docs/*.md` (17)	Write-ups: dataset validation, loopback, transitions, workflow discovery, pathway variation, waiting time, activity frequency, distribution, journey sequence, journey transition, bottleneck, performance summary, recommendations, visualization dashboard, interactive dashboard, journey comparison, project overview
`docs/*.png` (3)	activity*frequency*chart, patient*pathway*chart, workflow*elapsed*time_chart
`docs/*.svg` (2)	patient*workflow*map (full log), process_map (3-case prototype)
`data/raw/sample_event_log.csv`	16-event prototype log
`data/processed/` (20 CSV)	activity*distribution*summary, activity*elapsed*times, activity*frequency*summary, activity*transition*details, activity*transition*summary, loopback*activity*summary, loopback*case*summary, loopback*events, pathway*variation*summary, patient*journey*comparison*summary, patient*journey*sequence*summary, patient*journey*summary, patient*journey*transition*summary, patient*waiting*time*summary, process*bottleneck*summary, repeat*activity*summary, transition*waiting*summary, workflow*improvement*recommendations, workflow*performance*summary, workflow*visualization*pathway*details

A.3 Branch sagar

File group	Contents
`notebooks/` (14)	N-03 to N-16 (Chapter 9)
`data/processed_new/` (16)	Case and event tables plus audit, outlier, segmented, peak-load, importance, executive and final-metrics artefacts
`data/processed_old/` (4)	Legacy datasets (for example `cleaned_careflow.csv`) retained for reference
`dashboard/` (4 + notes)	Four `.pbix` reports and `dashboard_p4_notes.md`
Root notes	`README.md`, `STAKEHOLDER_SUMMARY.md`, `day18_notes.md`

Appendix B Pathway Definitions

Complete activity sequences of the six variants, with the number of cases and the share of the training and test cohorts.

V1	267 cases, mean 58.5 min (train 219, test 48)
Sequence	Registration → Triage → Doctor Assessment → Discharge
Length	4 activities

V2	216 cases, mean 102.3 min (train 166, test 50)
Sequence	Registration → Triage → Doctor Assessment → X-Ray → Radiology Result → Documentation Review → Discharge
Length	7 activities

V3	149 cases, mean 102.9 min (train 115, test 34)
Sequence	Registration → Triage → Doctor Assessment → Lab Collection → Lab Result → Discharge
Length	6 activities

V4	146 cases, mean 132.8 min (train 117, test 29)
Sequence	Registration → Triage → Doctor Assessment → X-Ray → Radiology Result → Triage → Documentation Review → Doctor Reassessment → Discharge
Length	9 activities

V5	133 cases, mean 146.6 min (train 111, test 22)
Sequence	Registration → Triage → Doctor Assessment → X-Ray → Radiology Result → Documentation Review → Lab Collection → Lab Result → Discharge
Length	9 activities

V6	89 cases, mean 174.6 min (train 72, test 17)
Sequence	Registration → Triage → Doctor Assessment → X-Ray → Radiology Result → Triage → Documentation Review → Doctor Reassessment → Lab Collection → Lab Result → Discharge
Length	11 activities

Appendix C Transition Matrix

Counts of directly-follows pairs. Rows are the "from" activity, columns the "to" activity. Zero cells are shown as a dash.

From \ To	Reg	Tri	DAs	XRy	RRs	DRv	DRe	LCo	LRs	Dis
Registration	-	1,000	-	-	-	-	-	-	-	-
Triage	-	-	1,000	-	-	235	-	-	-	-
Doctor Assessment	-	-	-	584	-	-	-	149	-	267
X-Ray	-	-	-	-	584	-	-	-	-	-
Radiology Result	-	235	-	-	-	349	-	-	-	-
Documentation Review	-	-	-	-	-	-	235	133	-	216
Doctor Reassessment	-	-	-	-	-	-	-	89	-	146
Lab Collection	-	-	-	-	-	-	-	-	371	-
Lab Result	-	-	-	-	-	-	-	-	-	371
Discharge	-	-	-	-	-	-	-	-	-	-

Abbreviations: Reg Registration, Tri Triage, DAs Doctor Assessment, XRy X-Ray, RRs Radiology Result, DRv Documentation Review, DRe Doctor Reassessment, LCo Lab Collection, LRs Lab Result, Dis Discharge.

Appendix D Glossary

Term	Meaning
Acuity / ESI	Emergency Severity Index, 1 (most urgent) to 5 (least urgent).
Boarding / midnight crossing	A case that ends after midnight of its arrival day.
Case	One patient encounter from Registration to Discharge.
Cohort	A group of cases defined by a shared property, for example critical patients.
Data leakage	Use of information in training or prediction that would not be available at prediction time.
DFG	Directly-follows graph of an event log.
Door to doctor	Minutes from Registration to Doctor Assessment.
Elapsed time	Minutes between the timestamps of consecutive events.
Event log	Table of case, activity and timestamp records.
Feature store	Table of model inputs and targets prepared in advance (here `careflow_model_ready_scaled.csv`).
IQR	Interquartile range, Q3 minus Q1; used in the outlier rule Q1 - 1.5 IQR, Q3 + 1.5 IQR.
Length of stay (LOS)	Minutes from the first to the last recorded event of a case.
Loopback / rework	Return to an earlier stage and repetition of steps.
Min-Max scaling	Linear scaling to the range 0 to 1 using training minimum and maximum.
NRMSE	Root mean squared error divided by the standard deviation of the true values.
Process mining	Techniques that discover and analyse processes from event logs.
Structural missingness	A value absent because the event did not occur, not because it was lost.
Trace / variant	Ordered activity list of a case / a distinct trace.
Z-score	Value minus training mean, divided by training standard deviation.

Abbreviations

Abbreviation	Expansion
ED	Emergency Department
EDA	Exploratory data analysis
ESI	Emergency Severity Index
KPI	Key performance indicator
LOS	Length of stay
ML	Machine learning

Abbreviation	Expansion
PBIX	Power BI report file
PM4Py	Process Mining for Python library
NRMSE	Normalised root mean squared error
PR-AUC	Area under the precision-recall curve
SVG	Scalable Vector Graphics